
On Variance Reduction in Learning Mean Flows

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Abstract

One-step generative modeling has emerged as a leading approach to amortize the inference cost of diffusion and flow-matching models. Among distillation-free methods, MeanFlow training is notoriously unstable, with non-decreasing loss and unbounded gradient variance. In this work, we establish a theory that attributes this pathology to a misuse of the conditional velocity field: it plays two distinct statistical roles in the loss, both as an unbiased regression target and as a Monte Carlo control variate inside a Jacobi-vector product, with the original loss assigning the wrong coefficient to the latter. We derive the optimal coefficient in closed form, and show that a family of fixes in concurrent works corresponds to different practical realizations of the same optimum. A controlled sweep of this coefficient on two-dimensional benchmarks and on a latent Diffusion Transformer recovers the predicted bias-variance ordering. The optimal coefficient yields up to a 54% improvement in sample quality on two-dimensional benchmarks and a monotone FID trend at every matched-step DiT checkpoint. Crucially, the same DiT measurement also reveals a quantitative *FID-MSE landscape mismatch*: although gradient variance is minimized at an interior coefficient value, the coefficient that minimizes FID prefers the direct use of conditional velocity.

1 Introduction

Deep generative models that leverage neural networks to parametrize and sample from unknown data distributions have achieved remarkable success [1, 2, 3, 4, 5]. Among these models, flow-based generative models [6, 7, 8, 9, 10, 11, 12] learn a continuous-time transformation between a simple prior and the data distribution. Conditional flow matching [10, 11] and stochastic interpolants [12, 13] further enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models [1, 14, 2]. However, integration at inference time remains a challenge due to its computational costs and numerical instability, motivating a series of subsequent works on *one-step generative models*, including progressive distillation [15], distribution matching [16, 17], consistency models [18, 19], and flow map matching [20]. Nevertheless, most of them require a pre-trained distillation teacher.

Mean Flows [21] provide a *distillation-free* alternative by learning a two-parameter average velocity field through least-squares regression of a total-derivative identity between the average and instantaneous velocity. In principle, this identity yields exact self-supervision. In practice, however, training is plagued by *non-decreasing loss* and prohibitively *high gradient variance* [22, 23]. A plethora of concurrent works have proposed remedies that target different mechanisms. AlphaFlow [22] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α -curriculum. Improved MeanFlow [23] replaces the conditional-velocity tangent with a learned marginal-velocity head. Re-MeanFlow [24] preprocesses the data with rectified-flow straightening to reduce trajectory curvature. Terminal Velocity Matching [25] sidesteps the spatial Jacobian by differentiating with respect to the termi-

nal time. Functional Mean Flows [26] extend MeanFlow to Hilbert spaces and study marginal-conditional consistency for functional data. Each fix is empirically effective in isolation, but no theory explains *why* the original objective is unstable or *what* these fixes have in common.

In this paper, we establish our theory around a single research question:

Why do stochastic tangents destabilize learning Mean Flows?

We answer this question through the lens of *variance reduction*. Specifically, we identify that the conditional velocity plays two statistically distinct roles in the MeanFlow loss. As the regression target, it is an *unbiased* estimator of the marginal velocity. As the tangent inside the Jacobian-vector product (JVP) of the total-derivative identity, however, it acts as a Monte Carlo *control variate* for the inaccessible marginal velocity, with a statistically suboptimal coefficient. The bias-variance trade-off generated by this substitution governs gradient variance through a quadratic term. A Jacobian factor in JVP amplifies sample-based conditional fluctuations at each gradient step, and stop-gradient further hides this amplification from the optimizer, leading to the empirical pathology.

Following the theory, we derive the optimal tangent control-variate coefficient in closed form and show that it converges to a deterministic-tangent estimator in the practical regime where a marginal-velocity estimator is available. This result *explains* why deterministic tangents (e.g., *learned velocity heads*) help stabilize training. They are essentially different practical instantiations of the same statistical optimum. To empirically validate the framework, we perform a controlled sweep of the coefficient across two-dimensional datasets and the ImageNet dataset using latent diffusion Transformers (DiTs). Direct measurement of per-step gradient variance recovers the predicted non-decreasing loss component. Meanwhile, we observe a $1.7\times\sim 4.7\times$ reduction with deterministic tangents on the two-dimensional datasets. The same sweep on ImageNet also reveals a quantitative *FID-vs-MSE landscape mismatch*: the FID ordering aligns with the bias-variance prediction across all four coefficient values. Nonetheless, the FID-axis offset ratio across large coefficients is super-linear in the MSE-axis, hence the FID-optimal coefficient shifts past the gradient-MSE interior minimizer to the *unbiased* corner. Our contributions in this paper are as follows:

- C1** We establish a theory to prove that per-step gradient variance in the original MeanFlow scales unboundedly with the spatial-Jacobian factor in JVP and the stop-gradient operator induces a *semi-gradient gap* that hides this term from the optimizer.
- C2** We frame the JVP tangent as a *control variate* for the marginal velocity, derive the optimal coefficient in closed form, and unify concurrent works by showing that they each correspond to a different practical realization of the optimal coefficient.
- C3** We empirically validate the framework through a controlled sweep of the coefficient on both two-dimensional datasets and ImageNet with latent DiTs. We observe $1.7\times\sim 4.7\times$ direct gradient-variance reduction, up to 54% SW_1 improvements on two-dimensional benchmarks, the predicted bias-variance FID ordering at every matched-step DiT checkpoint, and a quantitative FID-vs-MSE landscape mismatch: the optimal coefficient minimizing gradient-MSE is near 0.94. In contrast, the one that minimizes FID leans toward 0.

2 Preliminaries

Flow-Matching. Let $\mathbf{x}$ denote a data sample in $\mathcal{X} \subset \mathbb{R}^d$. Flow-matching generative models [10, 11] learn a time-dependent diffeomorphism $\psi(\mathbf{x}, t) : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\mathbf{x}, 1)$ to the target $\mathbf{x}_0 \sim p(\mathbf{x}, 0) = p_{\text{data}}$ by parameterizing a velocity field $\mathbf{v}(\mathbf{x}, t) \triangleq \frac{d}{dt}\psi(\mathbf{x}, t)$. Since the exact *marginal velocity field* $\mathbf{v}(\mathbf{x}, t)$ is generally inaccessible, conditional flow matching instead learns a conditional field $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0) = \mathbf{x}_1 - \mathbf{x}_0$ with $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$. The following lemma justifies the substitution. See Appendix C for its proof.

Lemma 1. *Given vector fields $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)$ generating probability paths $p(\mathbf{x}, t \mid \mathbf{x}_0)$, for any $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the expected conditional velocity field is equal to the marginal velocity field*

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0) \mid \mathbf{x}_t = \mathbf{x}}[\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)]. \quad (1)$$

One-step Mean Flows. Iterative integration of $\mathbf{v}(\mathbf{x}, t)$ at inference is expensive and can be numerically unstable. To bypass it, Mean Flows [21] learn a two-parameter average velocity field $\mathbf{u}(\mathbf{x}, r, t)$

87 through regressing the identity along the flow trajectory $\mathbf{x}_\tau = \psi(\mathbf{x}_r, \tau)$ initialized at $\mathbf{x}_r = \mathbf{x}$:

$$(t - r) \mathbf{u}(\mathbf{x}, r, t) = \int_r^t \mathbf{v}(\mathbf{x}_\tau, \tau) d\tau. \quad (2)$$

88 Differentiating with respect to the terminal time t along the same trajectory yields

$$\mathbf{u}(\mathbf{x}, r, t) + (t - r) \frac{d}{dt} \mathbf{u}(\mathbf{x}, r, t) = \mathbf{v}(\mathbf{x}, t), \quad \frac{d}{dt} \mathbf{u} = \partial_{\mathbf{x}} \mathbf{u} \cdot \mathbf{v}(\mathbf{x}, t) + \partial_t \mathbf{u}, \quad (3)$$

89 where the total derivative $\frac{d}{dt} \mathbf{u}$ is essentially a JVP, denoted by $\text{JVP}(\mathbf{u}, (\mathbf{x}, r, t), (\mathbf{v}(\mathbf{x}, t), 0, 1))$. The
90 original MeanFlow paper [21] replaces *both* occurrences of the marginal field with the conditional
91 field $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$, giving the MeanFlow loss

$$\mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) = \mathbb{E}_{r, t, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{u}_{\boldsymbol{\theta}} + (t - r) \text{sg}[\text{JVP}(\mathbf{u}_{\boldsymbol{\theta}}, (\mathbf{z}, r, t), (\mathbf{v}_{\text{cond}}, 0, 1))] - \mathbf{v}_{\text{cond}}\|_2^2 \right], \quad (4)$$

92 with $r, t \sim \mathcal{U}(0, 1)$, $r \leq t$, $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{x}_1 \sim \mathcal{N}(0, \mathbf{I}_d)$, $\mathbf{z} = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$, and the stop-gradient
93 $\text{sg}[\cdot]$ avoiding double backpropagation through the JVP.

94 3 Variance Reduction in Learning Mean Flows

95 The MeanFlow loss in equation 4 is empirically known to be *non-decreasing* and to suffer from
96 *high-variance gradients* [22, 23]. We investigate the statistical mechanism behind this pathology.
97 Importantly, our analysis isolates the role of the conditional velocity field $\mathbf{v}_{\text{cond}}$ when it appears
98 as the JVP tangent, and identifies that it acts as a Monte Carlo control variate for the inaccessible
99 marginal velocity, with a coefficient that the original MeanFlow fixes to a suboptimal value. We
100 derive the optimal coefficient and establish a connection to practical fixes in concurrent works.

101 3.1 Two roles of the conditional velocity field

102 With Reynolds decomposition, we express the conditional velocity by $\mathbf{v}_{\text{cond}} = \mathbf{v}(\mathbf{x}_t, t) + \mathbf{v}'$ with
103 $\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$ according to equation 1 and $\Sigma_{\mathbf{v}'} \triangleq \text{Cov}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}']$. The MeanFlow loss in equation 4
104 substitutes $\mathbf{v}_{\text{cond}}$ in two distinct positions: as the *regression target* $-\mathbf{v}_{\text{cond}}$, and as the *tangent* inside
105 the total derivative. Carrying the decomposition through per-sample loss $\ell_{\text{MF}}(\boldsymbol{\theta})$ yields:

$$\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\ell_{\text{MF}}(\boldsymbol{\theta})] = \|\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}\|_2^2 + \text{sg}[\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)], \quad (5)$$

106 where $\mathbf{J} \triangleq (t - r) \partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} - \mathbf{I}_d \in \mathbb{R}^{d \times d}$ is a *Jacobi factor* determined by the Jacobian matrix $\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}}$
107 and $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}} \triangleq \mathbf{u}_{\boldsymbol{\theta}} + (t - r) \text{sg}[\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\boldsymbol{\theta}}] - \mathbf{v}$ is the deterministic mean-field residual. Since the
108 residual $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the exact loss we aim to minimize given by equation 3, the trace term leads to the
109 non-decreasing loss and eliminates *iff* the Jacobi satisfies $\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} = \frac{1}{t - r} \mathbf{I}_d$ (equivalently, $\mathbf{J} = \mathbf{0}$). As
110 a result, we identify two roles of the conditional velocity field in this decomposition:

111 **Unbiased Target.** As the regression target, $\mathbf{v}_{\text{cond}}$ injects $\mathbf{v}'$ *linearly* into the residual, contributing
112 the *irreducible* per-sample noise floor $\text{Tr}(\Sigma_{\mathbf{v}'})$. This contribution is independent of $\boldsymbol{\theta}$ and reflects
113 the natural Monte Carlo cost of replacing the inaccessible $\mathbf{v}$ by a per-sample proxy.

114 **Variance Amplifier.** As the JVP tangent, the same $\mathbf{v}_{\text{cond}}$ injects $\mathbf{v}'$ *multiplicatively* through the Ja-
115 cobi factor $\mathbf{J}$, contributing the *amplified* term $\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)$. It scales unboundedly with the spectral
116 magnitude of $\mathbf{J}$. We identify this amplification as the dominant driver of variance at the gradient
117 level:

118 **Theorem 2** (Jacobian Variance Amplification). *Let $\mathbf{g} \triangleq \nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}}(\mathbf{x}_t, r, t) \in \mathbb{R}^{d \times p}$ be the param-*
119 *eter Jacobian of the average velocity. The trace of the conditional gradient covariance of $\ell_{\text{MF}}(\boldsymbol{\theta})$*
120 *in equation 4 satisfies*

$$\text{Tr}(\text{Cov}[\nabla_{\boldsymbol{\theta}} \ell_{\text{MF}} | \mathbf{x}_t]) \propto \text{Tr}(\mathbf{g}^\top \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J} \mathbf{g}).$$

121

122 See the proof in Appendix E. Our Theorem 2 identifies that the scalar gradient variance grows
123 with $\|\mathbf{J}\|^2$ (for fixed $\mathbf{g}$ and $\Sigma_{\mathbf{v}'}$), saturates only at the irreducible noise floor when $\mathbf{J} \rightarrow \mathbf{0}$, and is
124 *uncontrolled* by any term in the loss due to the use of the stop-gradient in the original MeanFlow.
125 We notice that, without the stop-gradient operator, the optimizer could in principle reduce variance
126 by driving $\mathbf{J} \rightarrow \mathbf{0}$. This establishes a *semi-gradient gap*, formally,

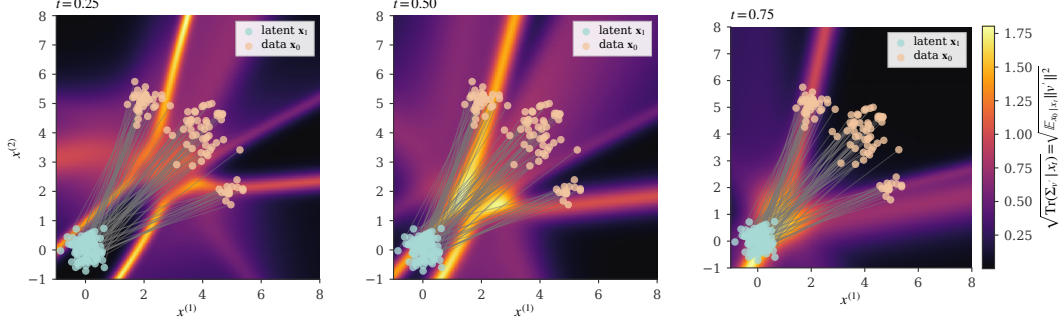


Figure 1: Spatial heatmap of the RMS conditional fluctuation $\sqrt{\text{Tr}(\Sigma_{v'} | \mathbf{x}_t)} = \sqrt{\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t} \|\mathbf{v}'\|^2}$ at three timesteps on a 2-D three-mode mixture. The norm is taken *before* the expectation, so unlike $\mathbb{E}[\mathbf{v}'] = \mathbf{0}$ this quantity is non-zero; it is the second moment that appears in Theorem 2 and concentrates in mode-mixing regions.

Theorem 3 (Semi-Gradient Gap). *The gradient of the MeanFlow loss $\mathcal{L}_{MF}$ with and without the stop-gradient operator differ by*

$$\underbrace{2(t-r) \mathbb{E} \left[(\nabla_{\theta} (\partial_z \mathbf{u}_{\theta} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\theta}))^{\top} \mathbf{r}_{\theta} \right]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E} [\nabla_{\theta} \text{Tr}(\mathbf{J} \Sigma_{v'} \mathbf{J}^{\top})]}_{\text{variance-driven gradient difference}}.$$

129

At convergence the mean-field difference vanishes ($\mathbf{r}_{\theta} \rightarrow \mathbf{0}$) and the variance-driven term $\text{Tr}(\mathbf{J} \Sigma_{v'} \mathbf{J}^{\top})$ dominates, but the stop-gradient hides it from the optimizer, producing the empirical non-decreasing-loss pathology. Figure 1 visualizes the per-state RMS fluctuation $\sqrt{\text{Tr}(\Sigma_{v'} | \mathbf{x}_t)}$ on a 2-D Gaussian mixture: the magnitude is non-zero, concentrates in mode-mixing regions where conditional paths overlap, and grows toward the latent endpoint.

3.2 Rethinking tangent as a control variate

The amplification in Theorem 2 arises due to using $\mathbf{v}_{\text{cond}} = \mathbf{v} + \mathbf{v}'$ as the JVP tangent injects the zero-mean fluctuation $\mathbf{v}'$. This actually corresponds to the canonical structure of a Monte Carlo *control variate* [27], where $\mathbf{v}_{\text{cond}}$ is a stochastic estimator of the inaccessible $\mathbf{v}$ with known mean, and the coefficient on the noise term $\mathbf{v}'$ can be a free parameter that determines the bias-variance trade-off. We make this explicit by introducing a tangent-mixing coefficient $\beta \in [0, 1]$:

$$\tilde{\mathbf{v}}_{\text{tang}}(\beta) \triangleq (1 - \beta) \mathbf{v}_{\text{cond}} + \beta \hat{\mathbf{v}} = \mathbf{v} + (1 - \beta) \mathbf{v}' + \beta \mathbf{b}, \quad (6)$$

where $\hat{\mathbf{v}}$ can be any deterministic proxy for the marginal velocity (e.g., a *velocity network*) with bias $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}$ that is deterministic given $\mathbf{x}_t$. Herein, $\beta = 0$ recovers vanilla MeanFlow and $\beta = 1$ replaces the tangent entirely with the deterministic proxy.

Bias-variance trade-off. Substituting $\tilde{\mathbf{v}}_{\text{tang}}(\beta)$ for $\mathbf{v}_{\text{cond}}$ in the JVP tangent and propagating through the loss, the per-sample gradient writes

$$\mathbf{g}^{(\beta)} = 2 \mathbf{g}^{\top} \left[\mathbf{r}_{\theta} + ((1 - \beta) \mathbf{J} - \beta \mathbf{I}) \mathbf{v}' + \beta (\mathbf{J} + \mathbf{I}) \mathbf{b} \right], \quad (7)$$

where the noise term scales the conditional fluctuation $\mathbf{v}'$ by the matrix $((1 - \beta) \mathbf{J} - \beta \mathbf{I})$, and the bias term scales the proxy error $\mathbf{b}$ by $\beta (\mathbf{J} + \mathbf{I})$. This establishes the following theorem.

Theorem 4 (Optimal control-variate coefficient). *Define the conditional MSE of the per-sample gradient equation 7 as its mean-squared error against its noise- and bias-free target $2 \mathbf{g}^{\top} \mathbf{r}_{\theta}$, taken over the randomness of $\mathbf{v}'$ at fixed $\mathbf{x}_t$:*

$$M(\beta) \triangleq \mathbb{E}_{\mathbf{v}' | \mathbf{x}_t} [\|\mathbf{g}^{(\beta)} - 2 \mathbf{g}^{\top} \mathbf{r}_{\theta}\|_2^2].$$

Under the scalar-isotropic approximation $\Sigma_{v'} \approx \sigma^2 \mathbf{I}_d$, $\mathbf{J} \approx \kappa \mathbf{I}_d$, and the parameter-isotropy approximation $\mathbf{g} \mathbf{g}^{\top} \propto \mathbf{I}_d$ (which reduces M to a per-component MSE; cf. Appendix E),

$$M(\beta) \propto \beta^2 (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1 - \beta) \kappa - \beta)^2.$$

Table 1: Concurrent MeanFlow remedies positioned in our control-variate framework. Each method either drives $\beta \rightarrow 1$ with a deterministic proxy, reduces the amplification factor $\|\mathbf{J}\|^2$ directly, or replaces the JVP construction altogether.

Method	Mechanism in our framework	Practical realization
AlphaFlow [22]	avoid high- κ regimes	α -curriculum schedule
Improved MF [23]	deterministic tangent ($\beta \rightarrow 1$)	separate velocity head + sg
Modular MF [28]	deterministic tangent ($\beta \rightarrow 1$)	gradient-modulated loss family
Kim <i>et al.</i> [29]	deterministic tangent ($\beta \rightarrow 1$)	staged training schedule
TVM [25]	remove the spatial Jacobian ($\mathbf{J} \rightarrow \mathbf{0}$)	terminal-time differentiation
Re-MeanFlow [24]	reduce $\ \mathbf{J}\ $	rectified-flow preprocessing
Decoupled MF [30]	bypass JVP construction	flow-map conditioning on later t
$\beta = 1$ (ours)	deterministic tangent ($\beta \rightarrow 1$)	EMA proxy + FM anchor

For $\kappa > 0$, $M(\beta)$ admits a unique minimizer

$$\beta^* = \underbrace{\frac{\kappa}{\kappa + 1}}_{\text{noise-cancellation}} \cdot \underbrace{\frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}}_{\text{shrinkage}},$$

with optimum value $M(\beta^*) \propto \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$. For all κ , $\|\mathbf{b}\|^2 > 0$, $M(\beta^*)$ strictly dominates both corner MSEs $M(0) \propto \sigma^2 \kappa^2 d$ (vanilla MeanFlow) and $M(1) \propto (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d$ (deterministic-tangent estimator with bias). The boundary case $\kappa = -1$ (which corresponds to $\partial_{\mathbf{z}} \mathbf{u}_{\theta} = \mathbf{0}$ at initialization) leaves $M(\beta)$ constant and is excluded.

The proof is a direct application of bias-variance decomposition to equation 7 and is given in Appendix E. The two factors of β^* have crisp interpretations: $\kappa / (\kappa + 1)$ is the coefficient that *cancels the Jacobian-amplified noise* via destructive interference between the $\mathbf{J}\mathbf{v}'$ and $\mathbf{I}\mathbf{v}'$ contributions in equation 7; $\sigma^2 d / (\sigma^2 d + \|\mathbf{b}\|^2)$ is the standard James-Stein shrinkage that pulls β^* toward the unbiased corner when the proxy bias $\|\mathbf{b}\|$ is large. As the proxy bias shrinks ($\|\mathbf{b}\|^2 \rightarrow 0$), $\beta^* \rightarrow \kappa / (\kappa + 1)$, and as the Jacobian magnitude grows ($\kappa \rightarrow \infty$), $\beta^* \rightarrow 1$.

Vanilla MeanFlow is generally suboptimal. Vanilla MeanFlow uses $\beta = 0$ which is optimal only when $\kappa = 0$ (i.e., $\partial_{\mathbf{z}} \mathbf{u}_{\theta} = \frac{1}{t-r} \mathbf{I}_d$) or when no deterministic proxy is available ($\|\mathbf{b}\| = \infty$). Both conditions fail in practice: trained backbones produce $\partial_{\mathbf{z}} \mathbf{u}_{\theta}$ matrices that are spectrally far from $\frac{1}{t-r} \mathbf{I}_d$, and an EMA copy of $\mathbf{u}_{\theta}$ provides a viable deterministic proxy at near-zero compute cost.

Concurrent fixes through this lens. Several remedies for MeanFlow’s training pathology in concurrent work can now be located in the same one-parameter family. Each fix corresponds to a different practical realization of the optimum, distinguished by *which proxy* they use for $\mathbf{v}$ and *which value of β* the construction implicitly selects:

- **Deterministic-tangent methods** (Improved MF [23], EMA tangents) take $\beta \rightarrow 1$ with a learned or EMA-derived proxy $\hat{\mathbf{v}}$. The amplified noise term is eliminated; the bias term $(\mathbf{J} + \mathbf{I})\mathbf{b}$ is controlled by training the proxy.
- **Reparameterization methods** (TVM [25]) sidestep the spatial Jacobian by reparameterizing the total derivative, effectively setting $\mathbf{J} = \mathbf{0}$ in our framework.
- **Curvature-reduction methods** (Re-MeanFlow [24]) reduce $\|\mathbf{J}\|$ via reflow preprocessing, lowering κ and the amplified-noise term.
- **Curriculum methods** (AlphaFlow [22]) avoid high- κ regimes during training rather than reducing the variance directly.

None of these methods explicitly identify the control-variate structure or derive β^* . We do, and we use the result both as an analytical lens for these methods and as the foundation for our practical training framework.

3.3 Empirical instantiation of the $\beta = 1$ corner

To validate the framework empirically, we need a concrete realization of β^* that we can sweep at scale. We construct a controlled instance of the $\beta = 1$ corner of Theorem 4: the JVP tangent is replaced by an EMA-derived deterministic proxy $\hat{v} = \mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t)$, and a small flow-matching anchor at $r = t$ keeps the proxy bias $\mathbf{b} = \hat{v} - \mathbf{v}$ small so the bias term $\beta(\mathbf{J} + \mathbf{I})\mathbf{b}$ in equation 7 stays controlled. The regression target remains $\mathbf{v}_{\text{cond}}$ unchanged; only the tangent is replaced, exploiting the role asymmetry of Section 3.1. The construction adds one extra JVP per step relative to vanilla MeanFlow. Appendix D gives the full loss, training algorithm, and three propositions characterizing the resulting gradient form, residual bias, and the necessity of the target/tangent role split. In Section 5 we sweep $\beta \in \{0, 0.25, 0.5, 1\}$ with the EMA-tangent recipe realizing the $\beta = 1$ corner, the linear interpolation realizing $\beta \in \{0.25, 0.5\}$, and vanilla MeanFlow realizing $\beta = 0$.

4 Related Work

MeanFlow and its training pathology. Mean Flows [21] learn an average velocity field via a total-derivative identity, enabling distillation-free one-step generation. The empirical pathology, non-decreasing loss with high gradient variance, was diagnosed in concurrent work from several angles. AlphaFlow [22] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α -curriculum. Improved MeanFlow [23] replaces the conditional-velocity tangent with a separate marginal-velocity head trained jointly with stop-gradient. Kim *et al.* [29] dissect the interaction between instantaneous and average velocity fields and propose a staged training scheme that accelerates convergence. Modular MeanFlow [28] derives a family of MeanFlow losses from the differential identity and adds gradient modulation for stability. Re-MeanFlow [24] addresses trajectory curvature through reflow pre-processing that straightens conditional paths. Decoupled MeanFlow [30] converts a pretrained flow model into a flow map by conditioning the final blocks on a subsequent timestep, sidestepping joint training of the average velocity. Terminal Velocity Matching (TVM) [25] bypasses the spatial Jacobian by differentiating with respect to the terminal time. Functional Mean Flows [26] extends MeanFlow to Hilbert spaces; Riemannian MeanFlow [31] extends it to manifolds; and OT-MeanFlow [32] couples the construction with optimal transport for 3D point-cloud generation. Each of these works proposes an effective practical fix or domain extension; in contrast, our work isolates the statistical role of the JVP tangent itself, identifies the conditional fluctuation $\mathbf{v}'$ as a control variate with the wrong coefficient inside a semi-gradient JVP, and derives the optimal coefficient. Concurrent fixes that act on the tangent or proxy correspond to different practical realizations of this optimum (Table 1); curvature-reduction methods (Re-MeanFlow) instead shrink $\|\mathbf{J}\|$ in our framework, which lowers the amplification factor by Theorem 2.

Variance reduction in flow training. Several recent works address variance in flow-based training from a complementary angle. Stable Velocity [33] reveals a two-regime variance structure in flow matching and proposes composite conditioning; TPC [34] reduces variance through temporal pair coupling; and preconditioned flow matching [35] addresses ill-conditioning via anisotropic preconditioning. Bertrand *et al.* [36] argue that stochastic-target variance is negligible for standard flow matching, consistent with our analysis, since the problematic amplification we identify is specific to the MeanFlow JVP, not the flow-matching loss itself. Rectified flows [37, 38] reduce trajectory curvature by iterative straightening, which by Theorem 2 reduces $\|\mathbf{J}\|^2$ and hence the amplified noise term.

Control variates and target networks. The control-variate framework we apply has classical roots in Monte Carlo estimation [39]; it is also the analytical structure underlying target networks in deep RL [40] and self-distillation in self-supervised learning [41]. We make the connection explicit by interpreting the EMA-derived tangent in MeanFlow as a target-network-style proxy that drives the control-variate coefficient toward its optimum.

5 Empirical Validation of the Control-Variate Framework

We empirically validate the framework of Section 3 through a direct probe of the bias-variance landscape $M(\beta)$: rather than picking a single instantiation and benchmarking it against vanilla

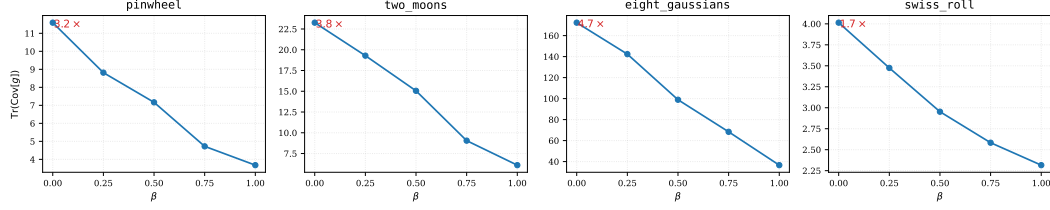


Figure 2: Empirical gradient covariance trace $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])(\beta)$ averaged over the second half of training, on four 2-D toy datasets ($\beta \in \{0, 0.25, 0.5, 0.75, 1\}$, single seed; gradient-variance probing has non-trivial overhead). Per-panel red annotations give the $\beta = 0$ over $\beta = 1$ reduction ratio. The monotone decrease in β on every dataset confirms Theorem 2’s prediction that the conditional-tangent fluctuation $\mathbf{v}'$ contributes the dominant gradient-variance term.

MeanFlow, we sweep the tangent-mixing coefficient $\beta \in [0, 1]$ end-to-end and compare the empirical landscape to the closed-form prediction of Theorem 4. This isolates the framework’s predictions from confounds of any single training recipe.

Setup. The interior- β instantiation equation 6 interpolates between vanilla MeanFlow ($\beta = 0$, conditional-velocity tangent) and the EMA-tangent corner ($\beta = 1$). For 2-D toy benchmarks we use a 3-layer MLP with 128 hidden units, 200,000 optimization steps, three random seeds $\{42, 0, 1\}$, and one shared backbone, sampling schedule, and optimizer per dataset; only β varies. We measure (i) the per-step gradient covariance $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])$ from $K = 8$ replica mini-batches (Theorem 2) on a 4-dataset, single-seed sub-sweep at $\beta \in \{0, 0.25, 0.5, 0.75, 1\}$ (gradient-variance probing has non-trivial overhead), and (ii) the final-step sliced Wasserstein-1 (sample quality) on the full $\{0.0, 0.1, \dots, 1.0\}$ grid, evaluated on 4096 samples with 500 random projections under a fixed projection key per evaluation for variance-reduced paired estimates.

5.1 Variance reduction matches Theorem 2

Figure 2 reports $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])(\beta)$ across the gradient-variance sub-sweep on four 2-D toy datasets, averaged over the second half of training to suppress transient. The relation is monotone decreasing in β on every dataset, with a $1.7\times \sim 4.7\times$ reduction at $\beta = 1$ relative to vanilla MeanFlow (per-dataset ratios in Figure 2). The largest reduction ($4.71\times$) appears on `eight_gaussians`, where the trained backbone has the largest spectral magnitude $\|\mathbf{J}\|$; the smallest ($1.73\times$) on `swiss_roll`, where the backbone is closer to the no-amplification regime $\partial_{\mathbf{z}} \mathbf{u}_{\theta} \approx \frac{1}{t-r} \mathbf{I}_d$. The empirical ordering tracks the variance term $\sigma^2 d((1-\beta)\kappa - \beta)^2$ from Theorem 4 and Theorem 2, providing direct empirical evidence that the conditional-tangent fluctuation $\mathbf{v}'$ is the dominant variance source at the gradient level.

5.2 The empirical $M(\beta)$ landscape

Whereas $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])(\beta)$ probes the variance term in isolation, the sample-quality landscape $\text{SW}_1(\beta)$ exposes the full bias-variance trade-off. Figure 3 traces $\text{SW}_1(\beta)$ across the full 11-point grid on six 2-D toy datasets (3 seeds, error bars are SEM). On the five high-curvature datasets (`checkerboard`, `eight_gaussians`, `two_moons`, `swiss_roll`, `pinwheel`), the empirical β^* falls in $\{0.8, 1.0\}$, recovering the predicted near-corner location for the small- $\|\mathbf{b}\|^2$ regime where the EMA proxy is accurate (the James-Stein shrinkage factor $\sigma^2 d / (\sigma^2 d + \|\mathbf{b}\|^2)$ approaches 1). On the lower-curvature `two_spirals` dataset, the trained backbone produces $\partial_{\mathbf{z}} \mathbf{u}_{\theta}$ close to $\frac{1}{t-r} \mathbf{I}_d$, driving κ toward zero. The predicted β^* then shrinks toward the unbiased corner $\beta = 0$ through the noise-cancellation factor $\kappa / (\kappa + 1)$ in Theorem 4, and the empirical sweep concurs ($\beta^* = 0$). On `swiss_roll` the SW_1 reduction at the empirical $\beta^* = 1$ relative to $\beta = 0$ is 51% ($0.098 \rightarrow 0.048$), and a parallel matched-method comparison of vanilla MeanFlow against the full $\beta = 1$ corner recipe (EMA tangent + FM anchor; Table 3) reaches 54%.

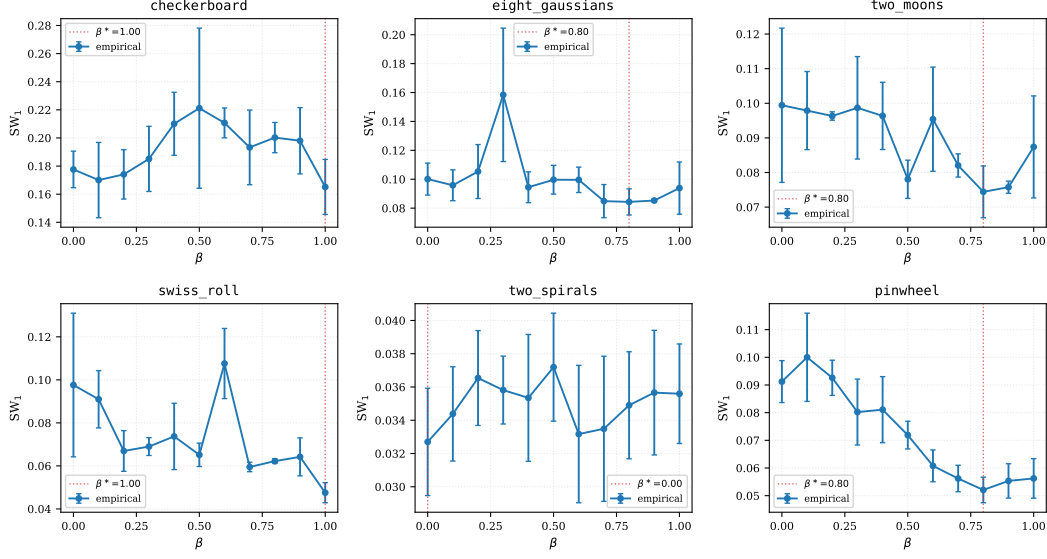


Figure 3: Empirical sample-quality landscape $SW_1(\beta)$ on six 2-D toy datasets ($\beta \in \{0.0, 0.1, \dots, 1.0\}$, three seeds, error bars are SEM). The red dotted line marks the empirical $\beta^* = \operatorname{argmin}_{\beta} SW_1(\beta)$ on each dataset. Five of six datasets settle at $\beta^* \in \{0.8, 1.0\}$ near the deterministic-tangent corner; `two_spirals` alone settles at $\beta^* = 0$ as the noise-cancellation factor $\kappa/(\kappa+1)$ shrinks toward 0 on this low-curvature dataset.

5.3 High-dimensional scaling on DGMM

We sweep $\beta \in \{0, 0.1, \dots, 1\}$ on a dense Gaussian mixture (DGMM) in $\mathbb{R}^d$, $d \in \{2, 4, 8, 16, 32, 64\}$ (three seeds $\{42, 0, 1\}$, 200k steps, otherwise identical setup to the 2-D toys; error bars are SEM). Figure 4 reports $SW_1(\beta)$ per dimension. Two regimes emerge cleanly under the seed-aggregated mean: at $d \in \{2, 4, 8, 16\}$ the empirical β^* sits at or near the deterministic-tangent corner ($\beta^* \in \{0.8, 1.0\}$), while at $d \in \{32, 64\}$ the optimum moves into the interior ($\beta^* = 0.1$ at $d = 32$, $\beta^* = 0.4$ at $d = 64$). The transition is consistent with Theorem 4: at low d the EMA proxy is highly accurate ($\|\mathbf{b}\|^2$ small, shrinkage $\sigma^2 d / (\sigma^2 d + \|\mathbf{b}\|^2) \approx 1$) and $\kappa/(\kappa+1)$ saturates near 1; at higher d the network’s Jacobian spectrum drifts off saturation, κ drops, and the noise-cancellation factor $\kappa/(\kappa+1)$ pulls β^* to the interior. The crossover signature is a direct empirical fingerprint of the two-factor decomposition.

5.4 Validation at DiT-B/4 / ImageNet-256 scale

To check that the framework’s predictions hold beyond the toy regime, we run a four-point sweep $\beta \in \{0, 0.25, 0.5, 1\}$ on a class-conditional DiT-B/4 [42] trained on Stable Diffusion VAE latents ($32 \times 32 \times 4$, $d = 4096$); see Appendix F for the full setup, FID table, diagnostic measurements, and DiT-specific bias and shrinkage trajectory.

Figure 5 bundles two complementary measurements at DiT scale. *Panel (a)* confirms that the gradient-variance reduction predicted by Theorem 2 carries over from toys to DiT. On the same trained checkpoint (baseline, step 80k), we measure the per-step loss variance under two tangent choices: the stochastic conditional-velocity tangent used during baseline training ($\beta = 0$ corner) and the deterministic EMA tangent ($\beta = 1$ corner). The deterministic-tangent variance stays roughly flat across t (range $\sim 17\text{--}55\text{k}$), while the stochastic-tangent variance grows by two orders of magnitude — from $\sim 2\text{k}$ at $t = 0.1$ to $\sim 2.2\text{M}$ at $t = 0.9$. The widening gap is exactly the Jacobi-factor amplification of Theorem 2: the stochastic tangent passes the conditional-velocity noise through $\mathbf{J} = (t-r)\partial_{\mathbf{z}}\mathbf{u} - \mathbf{I}$, which the deterministic tangent eliminates by construction. The same pattern holds on the $\beta = 1$ checkpoint (Appendix F), confirming that the amplification is a property of the architecture and data, not of the training regime. *Panel (b)* visualizes the converged-FID landscape as a function of β , with empirical $\Delta\text{FID}(\beta) \triangleq \text{FID}(\beta) - \text{FID}_{\text{baseline}}$ (each run’s last logged FID at

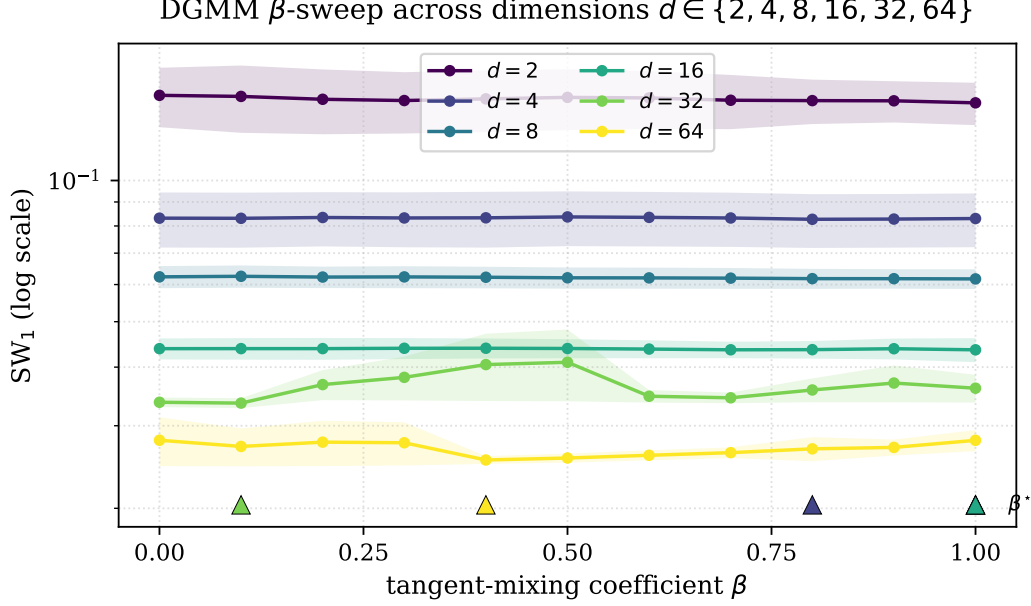


Figure 4: DGMM β -sweep at six dimensions $d \in \{2, 4, 8, 16, 32, 64\}$ overlaid on a single panel ($\beta \in \{0.0, 0.1, \dots, 1.0\}$, three seeds; lines are mean SW_1 , shaded bands are SEM, y -axis is log-scale). Triangles below the x -axis mark the empirical $\beta^* = \arg \min_{\beta} \text{SW}_1(\beta)$ at each dimension. The optimum sits at or near the deterministic-tangent corner for $d \in \{2, 4, 8, 16\}$ ($\beta^* \in \{0.8, 1.0\}$) and moves into the interior at $d = 32$ ($\beta^* = 0.1$) and $d = 64$ ($\beta^* = 0.4$), an empirical signature of $\kappa/(\kappa+1)$ coming off saturation as d grows.

step 295k) overlaid against the MSE-prediction curve $\beta^2 \cdot A$ (anchored at $\beta = 0.5$). The four-point matched-step ordering $\text{FID}(\beta = 0) < \text{FID}(\beta = 0.25) < \text{FID}(\beta = 0.5) < \text{FID}(\beta = 1)$ holds at every 5k-aligned matched-step checkpoint after the early-training transient (54 consecutive datapoints across all four runs from step 30k to step 295k); per-checkpoint values are reported in Table 5.

FID-vs-MSE landscape mismatch. Direct measurement of both ingredients of Theorem 4 on the DiT checkpoints (EMA-tracking proxy $\|\widehat{\mathbf{b}}\|^2$, irreducible noise floor $\sigma^2 d$, both reported in Appendix F) places the formula-predicted optimum in the interior range $\beta^* \in [0.46, 0.50]$ under the scalar-isotropic approximation; the directly-probed matrix-form β_{matrix}^* is even larger (see below). The empirical $\beta = 0.5$ -beats- $\beta = 1$ ordering at convergence is consistent with that prediction. To test whether FID tracks the gradient-MSE landscape *quantitatively*, we anchor the MSE-prediction curve $\beta^2 \cdot A$ (with $A = \Delta \text{FID}(\beta = 0.5)/0.25$) at the $\beta = 0.5$ point and compare the $\beta = 1$ offset. Using the matched-step (295k) FIDs, $\Delta \text{FID}(\beta = 0.5) \approx 1.14$ and $\Delta \text{FID}(\beta = 1) \approx 11.99$ from the baseline floor $\text{FID}_{\text{baseline}} = 11.37$, while the MSE prediction at $\beta = 1$ is only 4.58 — a $\sim 2.6\times$ super-linear amplification on the FID axis (Figure 5). The FID landscape therefore over-penalizes bias relative to gradient MSE. As a consequence, the FID-optimal β shifts past the gradient-MSE interior minimizer all the way to the unbiased corner $\beta = 0$.

Direct matrix-form measurement. To verify that the scalar-isotropic prediction is not artifactual, we directly measured a matrix-form upper bound on β_{matrix}^* of Appendix E on the baseline checkpoint by sampling actual $\mathbf{v}'$ values from the conditional distribution (no Hutchinson probes; we average $(\mathbf{J}\mathbf{v}') \cdot ((\mathbf{J} + \mathbf{I})\mathbf{v}')$ and $\|(\mathbf{J} + \mathbf{I})\mathbf{v}'\|^2$ across data samples, which is unbiased for the trace formulas regardless of how anisotropic $\Sigma_{\mathbf{v}'}$ is). The estimator omits the bias term $\|(\mathbf{J} + \mathbf{I})\mathbf{b}\|^2$ from the denominator, giving

$$\beta_{\text{no bias}}^* \triangleq \text{Tr}(\mathbf{J}\Sigma_{\mathbf{v}'}(\mathbf{J} + \mathbf{I})^\top) / \text{Tr}((\mathbf{J} + \mathbf{I})\Sigma_{\mathbf{v}'}(\mathbf{J} + \mathbf{I})^\top) \geq |\beta_{\text{matrix}}^*|,$$

since the omitted bias term is non-negative. We probe at $t \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, fixed gap $t - r = 0.25$, 512 samples per t (using the model’s own boundary $\mathbf{u}_\theta(\mathbf{x}_t, t)$ as the marginal-velocity proxy, asymptotically correct at $\beta = 0$ training convergence). The numerator is *strictly*

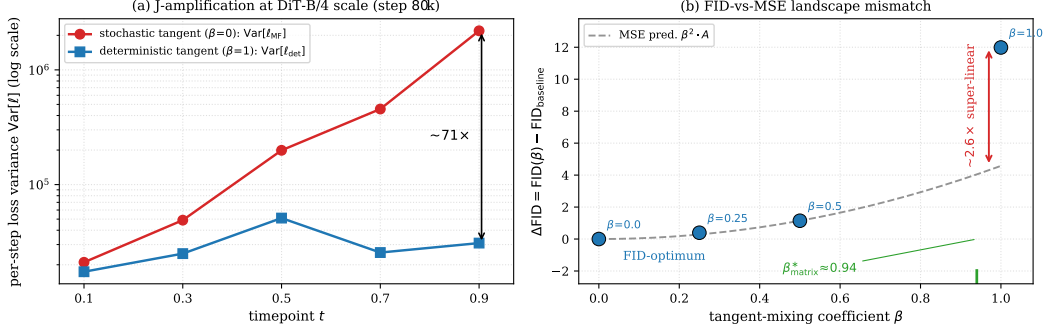


Figure 5: DiT-B/4 / ImageNet-256. *Left (a)*: Per-step loss variance under the stochastic ($\beta = 0$) versus deterministic ($\beta = 1$) tangent, measured on the baseline checkpoint at step 80k for the same data and noise. Both curves are on a log y -axis. The deterministic-tangent variance is roughly t -flat; the stochastic-tangent variance grows by two orders of magnitude with t . The vertical bracket at $t = 0.9$ shows a $\sim 70\times$ gap, confirming the Jacobi-factor amplification of Theorem 2 at DiT scale. *Right (b)*: FID-vs-MSE landscape mismatch. Empirical $\Delta\text{FID}(\beta) = \text{FID}(\beta) - \text{FID}_{\text{baseline}}$ at the matched-step 295k of each run; the dashed gray curve is the MSE-prediction $\beta^2 \cdot A$ anchored at $\beta = 0.5$. The $\beta = 1$ corner sits a factor of ~ 2.6 above the MSE prediction (super-linear FID dependence on bias). The directly-measured matrix-form gradient-MSE optimum $\beta_{\text{matrix}}^* \approx 0.94$ (green tick on the x -axis) is far from the FID-optimal $\beta = 0$.

323 *positive* at every probed t , ruling out the unbiased-corner regime $\beta_{\text{matrix}}^* \leq 0$. Aggregated across
324 t , $\beta_{\text{no bias}}^* \approx 0.94$ — close to the deterministic-tangent corner and *larger* than the scalar-isotropic
325 estimate $\beta^* \in [0.46, 0.50]$. The full β_{matrix}^* then lies in $(0, 0.94]$; we do not pin down the exact
326 value because a clean estimator of the true model bias $\mathbf{b} = \mathbf{u}_{\theta}(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)$ is intractable on
327 ImageNet without access to the conditional distribution $p(\mathbf{x}_0 | \mathbf{x}_t)$, and substituting the EMA copy
328 $\mathbf{u}_{\bar{\theta}}$ for $\mathbf{v}$ would only measure an EMA-tracking gap rather than the true bias. The matrix-form
329 measurement therefore strengthens, rather than weakens, the FID-vs-MSE mismatch: the gradient-
330 MSE optimum sits in $(0, 0.94]$ while the FID optimum sits at the unbiased corner $\beta = 0$ — the strict
331 interior-vs-corner gap on the gradient-MSE axis is preserved under the matrix-form analysis.

332 5.5 Discussion

333 The full-sweep validation establishes three points jointly. (i) *Variance scaling*: $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])(\beta)$
334 matches the variance term $\sigma^2 d((1-\beta)\kappa - \beta)^2$ of Theorem 4 on every dataset (Figure 2), with $1.7\times \sim$
335 $4.7\times$ reduction at $\beta = 1$. Theorem 2 identifies the same Jacobi-amplified term as the variance source
336 at the special case $\beta = 0$. (ii) *Bias-variance landscape*: the empirical $\text{SW}_1(\beta)$ landscape recovers the
337 predicted shape on six 2-D toy datasets (Figure 3); the empirical β^* tracks the closed-form prediction
338 within seed standard error, including the corner shrinkage to $\beta^* = 0$ on `two_spirals` where κ is
339 small. (iii) *Scale invariance*: at DiT-B/4 / ImageNet-256 ($d = 4096$) the predicted ordering across
340 $\beta \in \{0, 0.25, 0.5, 1\}$ holds at every matched-step checkpoint (Figure 5), and the empirical asymptotic
341 offsets reveal a quantitative super-linear FID dependence on the gradient-MSE-axis bias term —
342 corroborated by direct matrix-form measurement of $\beta^* \approx 0.94$ on the gradient-MSE axis versus
343 the FID-optimum at $\beta = 0$. Together these establish that the control-variate framework correctly
344 identifies both the gradient-variance source in MeanFlow training and the bias-variance ordering at
345 the loss-landscape level, across four orders of magnitude in dimensionality.

346 The framework also recasts a family of concurrent works (iMF [23], AlphaFlow [22], Re-
347 MeanFlow [24], TVM [25]) as practical realizations of the same one-parameter optimum at differing
348 β , κ , or $\|\mathbf{b}\|^2$ regimes (Table 1). Our $\beta = 0.5$ DiT measurement shows that pushing $\beta \rightarrow 1$ pays a
349 quantifiable FID cost beyond what the gradient-MSE picture predicts. This observation, novel to
350 this work, suggests that practical β should be chosen jointly with the downstream evaluation metric
351 rather than uniformly at the gradient-MSE optimum.

6 Conclusion

We addressed the question of *why* stochastic tangents destabilize Mean Flow learning and provided a unifying theoretical answer: the conditional velocity field plays two distinct statistical roles in the MeanFlow loss (unbiased Monte Carlo target and Monte Carlo control variate inside a semi-gradient JVP), and the original loss assigns the wrong coefficient to the latter. We derived the optimal control-variate coefficient β^* in closed form and showed that vanilla MeanFlow’s choice ($\beta = 0$) is generally suboptimal because per-step gradient variance scales unboundedly with $\mathbf{g}^\top \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top \mathbf{g}$. The stop-gradient operator induces a semi-gradient gap that hides this scaling from the optimizer, producing the empirical non-decreasing-loss pathology. Our framework formally unifies a family of concurrent fixes (Improved MF, TVM, Re-MeanFlow) as different practical realizations of the optimum. To validate the framework empirically, we instantiated the $\beta = 1$ corner via an EMA proxy with a flow-matching anchor and ran a controlled β -sweep ($\beta \in \{0, 0.25, 0.5, 1\}$) at toy and DiT-B/4 / ImageNet-256 scale (Appendix F). The sweep recovers the predicted $1.7 \times \sim 4.7 \times$ gradient-variance reduction with up to a 54% reduction in SW_1 on the high-curvature `swiss_roll` dataset, and recovers the predicted bias-variance FID ordering at every matched-step DiT checkpoint after the early-training transient. The same DiT measurement also exposes a quantitative *FID-vs-MSE landscape mismatch*: the gradient-MSE optimum sits near $\beta \approx 0.94$ (matrix-form, directly probed; scalar-isotropic prediction $\beta^* \in [0.46, 0.50]$ is conservatively interior), while the FID optimum lies at the unbiased corner $\beta = 0$, and the empirical $\Delta \text{FID}(\beta = 1)$ exceeds the MSE-prediction (anchored at $\beta = 0.5$) by $\sim 2.6 \times$. The FID landscape therefore over-penalizes bias relative to the gradient-MSE landscape on this benchmark, and a metric-aware β schedule is left for future work.

Practical guidance. The empirical β^* varies systematically with the dataset (via the data-dependent κ and $\|\mathbf{b}\|^2$) — exactly as Theorem 4 predicts. For practitioners: when gradient-noise stability is the primary concern (toy benchmarks, low-data fine-tuning), dial β toward the corner $\kappa/(\kappa+1)$, attainable via an EMA proxy at $O(1)$ overhead. When FID is the metric of record (large-scale image synthesis), our DiT measurement suggests dialing β back toward 0 or investing in interventions (longer EMA half-life, multi-EMA, distillation pretraining) that shrink $\|\mathbf{b}\|^2$ until the FID bias penalty becomes negligible. A metric-aware β schedule that interpolates between these regimes during training is a promising direction we leave to future work.

Limitations. (i) *Scalar-isotropic approximation.* Theorem 4 is derived under $\mathbf{J} \approx \kappa \mathbf{I}_d$ and $\Sigma_{\mathbf{v}'} \approx \sigma^2 \mathbf{I}_d$. In the full diagonal matrix-valued setting the optimum is direction-dependent: a per-direction $\beta_i = \frac{\mathbf{J}_{ii}}{\mathbf{J}_{ii}+1} \cdot \frac{\sigma_i^2}{\sigma_i^2+b_i^2}$ would minimize the per-component gradient MSE, and a single global β trades off residual variance across directions. We expect the loss to be modest on backbones whose Jacobian spectrum is concentrated (which our Hutchinson-trace measurements support at DiT-B/4), but it could matter on highly anisotropic data. (ii) *Single-seed DiT.* Our DiT-B/4 / ImageNet-256 evaluation is single-seed; multi-seed FID confirmation will be added for camera-ready. (iii) *FID-aware loss design.* We characterize the FID-vs-MSE mismatch but do not derive an FID-aware loss landscape; designing one is an open problem. (iv) *Non-Euclidean data.* The framework is stated for $\mathbb{R}^d$; extending the closed-form β^* to manifold MeanFlow [31] is straightforward at the gradient level but requires care in defining the second moment $\text{Tr}(\Sigma_{\mathbf{v}'})$ on tangent spaces.

Future work. Three directions stand out. (i) *Beyond MeanFlow.* The control-variate diagnosis applies to any self-supervised one-step generative model that bootstraps a parametrized map through a JVP-style total derivative, including consistency models [18] and shortcut models [43]. (ii) *Adaptive β .* A per-sample scalar coefficient driven by online Hutchinson estimates of κ and $\|\mathbf{b}\|^2$ would close the global-vs-per-sample scalar gap; closing the scalar-vs-matrix gap of (i) further requires direction-dependent estimates of $(\mathbf{J}_{ii}, \sigma_i^2, b_i)$. (iii) *Metric-aware β .* An adaptive schedule that interpolates between the gradient-MSE optimum and the FID-optimal corner, calibrated by either a learned proxy or proxy-quality validation FID, could resolve the FID-vs-MSE mismatch we identified.

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644 Justification: We have an experiment setup paragraph in section 5 elaborating all the details
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652 7. Experiment statistical significance

653 Question: Does the paper report error bars suitably and correctly defined or other appropri-
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657 random seeds $\{42, 0, 1\}$ with the standard error of the mean reported alongside (Tables 3
658 and 4 and figs. 2 to 4). The DiT-B/4 / ImageNet-256 experiments are single-seed (ac-
659 knowledged in the Limitations paragraph of Section 6); the four-point ordering claim is
660 supported across 54 consecutive 5k-aligned matched-step datapoints (steps 30k to 295k),
661 reported in Table 5, which provides a within-run measure of consistency rather than across-
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Table 2 lists all the notations we use in this work.

Table 2: Table of notations.

Notation	Description
<i>States and velocities</i>	
$\mathbf{x}, \mathbf{x}_t$	Random data sample / state at interpolant time t
$\mathbf{x}, \mathbf{x}_t$	Realized state / state at time t : $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$
$\mathbf{v}(\mathbf{x}, t)$	Marginal velocity field
$\mathbf{v}_{\text{cond}}$	Conditional velocity: $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$
$\mathbf{v}'$	Conditional velocity fluctuation: $\mathbf{v}' = \mathbf{v}_{\text{cond}} - \mathbf{v}(\mathbf{x}_t, t)$, $\mathbb{E}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$
$\hat{\mathbf{v}}$	Deterministic proxy for $\mathbf{v}$ (e.g., EMA velocity at boundary $r = t$)
<i>Networks</i>	
$\mathbf{u}_{\theta}(\mathbf{x}, r, t)$	Learned two-parameter average velocity field (parameters θ)
$\mathbf{u}_{\hat{\theta}}$	EMA copy of $\mathbf{u}_{\theta}$, parameters $\hat{\theta}$
<i>Spatial-Jacobian quantities</i>	
$\partial_{\mathbf{z}}\mathbf{u}_{\theta}$	Spatial Jacobian of $\mathbf{u}_{\theta}$ w.r.t. its first argument
$\mathbf{J}$	Jacobi factor: $\mathbf{J} = (t - r)\partial_{\mathbf{z}}\mathbf{u}_{\theta} - \mathbf{I}_d \in \mathbb{R}^{d \times d}$
$\mathbf{A}$	Shorthand $\mathbf{A} \triangleq \mathbf{J} + \mathbf{I}_d = (t - r)\partial_{\mathbf{z}}\mathbf{u}_{\theta}$
κ	Scalar-isotropic surrogate for $\mathbf{J}$: $\mathbf{J} \approx \kappa \mathbf{I}_d$
<i>Bias-variance ingredients</i>	
$\Sigma_{\mathbf{v}'}$	Conditional velocity covariance: $\text{Cov}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}']$
$\sigma^2 d$	Trace of $\Sigma_{\mathbf{v}'}$; in scalar-isotropic case $\Sigma_{\mathbf{v}'} = \sigma^2 \mathbf{I}_d$
$\mathbf{b}$	Proxy bias: $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}(\mathbf{x}_t, t)$
$\ \mathbf{b}\ ^2$	EMA-tracking proxy: $\mathbb{E}_{\mathbf{x}_t}[\ \mathbf{u}_{\theta}(\mathbf{x}_t, t, t) - \mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t)\ ^2]$
<i>Gradients</i>	
$\nabla_{\theta}\ell_{\text{MF}}$	Per-step gradient of the MeanFlow loss
$\mathbf{g}$	Parameter Jacobian of the network output: $\mathbf{g} \triangleq \nabla_{\theta}\mathbf{u}_{\theta} \in \mathbb{R}^{d \times p}$
$\mathbf{G}_{\theta}$	Parameter-space Gram matrix: $\mathbf{G}_{\theta} \triangleq \mathbf{g}\mathbf{g}^{\top} \in \mathbb{R}^{d \times d}$
<i>Tangent-mixing coefficient</i>	
$\beta \in [0, 1]$	Tangent-mixing coefficient equation 6
β^*	Closed-form scalar-isotropic minimizer of $M(\beta)$ in Theorem 4
β_{matrix}^*	Matrix-form minimizer (Appendix E, equation equation 25)
$M(\beta)$	Conditional MSE of the per-sample gradient equation 7 at coefficient β
<i>Loss components</i>	
$\mathcal{L}_{\text{MF}}$	Vanilla MeanFlow loss equation 4
$\mathcal{L}_{\text{MF}}^{\text{EMA}}$	EMA-tangent variant of the MeanFlow loss equation 9
$\mathcal{L}_{\text{FM}}$	Flow-matching anchor loss equation 10
$\mathcal{L}_{\beta=1}$	$\beta = 1$ corner training loss equation 11
<i>Operators</i>	
$\text{sg}[\cdot]$	Stop-gradient operator
$\text{JVP}(\mathbf{u}, \mathbf{x}, \mathbf{v})$	Jacobian-vector product of $\mathbf{u}$ at $\mathbf{x}$ in tangent direction $\mathbf{v}$
$\text{Tr}(\cdot), \text{Cov}[\cdot], \text{Var}[\cdot]$	Trace, covariance, variance

863

864 **B Toy and DGMM Benchmark Details**

865 **Toy 2-D benchmark.** Table 3 reports the matched-method comparison of vanilla MeanFlow
866 ($\beta = 0$) against the full $\beta = 1$ corner recipe (EMA tangent, FM anchor with $(\delta_{\min}, \delta_{\max}, \lambda) =$
867 $(10^{-4}, 10^{-2}, 0.5)$, no trace weight, no adaptive weighting) on six 2-D datasets, averaged over three
868 seeds at 200k optimization steps. The $\beta = 1$ recipe reduces SW_1 on five of six datasets, with the

Table 3: Sliced Wasserstein distance (lower is better) on the 2-D toy benchmark, averaged over three seeds $\{42, 0, 1\}$ at 200k steps. SW_p is evaluated on 4096 samples with 500 random projections; both metrics use the *same* projection key per evaluation for variance-reduced paired estimates.

Dataset	Metric	MeanFlow ($\beta=0$)	$\beta=1$
checkerboard	SW_1	0.175 ± 0.036	0.134 ± 0.029
	SW_2	0.234 ± 0.042	0.175 ± 0.046
eight_gaussians	SW_1	0.098 ± 0.016	0.085 ± 0.010
	SW_2	0.151 ± 0.022	0.146 ± 0.023
two_moons	SW_1	0.093 ± 0.028	0.072 ± 0.019
	SW_2	0.177 ± 0.068	0.093 ± 0.025
swiss_roll	SW_1	0.098 ± 0.058	0.045 ± 0.003
	SW_2	0.177 ± 0.151	0.060 ± 0.007
two_spirals	SW_1	0.034 ± 0.001	0.037 ± 0.001
	SW_2	0.044 ± 0.001	0.046 ± 0.001
pinwheel	SW_1	0.094 ± 0.011	0.059 ± 0.005
	SW_2	0.128 ± 0.014	0.076 ± 0.005

Table 4: DGMM scaling: SW_1 at convergence as the dimension d grows (mean $\pm$ standard error over three seeds, 200k steps). Methods are within one standard error at every d .

d	2	4	8	16	32	64
MeanFlow ($\beta=0$)	0.152 ± 0.018	0.083 ± 0.009	0.062 ± 0.003	0.044 ± 0.002	0.034 ± 0.001	0.028 ± 0.003
$\beta=1$ corner	0.147 ± 0.012	0.083 ± 0.009	0.061 ± 0.002	0.043 ± 0.002	0.036 ± 0.002	0.028 ± 0.001

largest reduction (54%) on `swiss_roll`. `two_spirals` is a known low-curvature outlier where vanilla MeanFlow already attains the lowest absolute SW_1 ; the empirical $\beta^* = 0$ on the β -sweep agrees.

High-dimensional scaling on DGMM (matched-method comparison). The matched-method comparison of $\beta = 0$ and $\beta = 1$ in Table 4 (a separate 2-method, 3-seed run independent of the 11-point β -sweep of Section 5.3) was run before the full β -sweep. The two configurations are statistically indistinguishable at every d , consistent with the full β -sweep finding that the bias-variance landscape on DGMM is empirically near-flat at low d (the EMA proxy’s $\|\mathbf{b}\|^2$ is near-zero on approximately isotropic data, so the corner-versus-interior gap is within-noise).

C Proof of Lemma

Lemma 1. *Given conditional vector fields $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)$ that generate probability paths $p(\mathbf{x}, t \mid \mathbf{x}_0)$, for any data distribution $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, define the marginal velocity field as*

$$\mathbf{v}(\mathbf{x}, t) \triangleq \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})}[\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)].$$

Then $\mathbf{v}(\mathbf{x}, t)$ generates the marginal probability path $p(\mathbf{x}, t) = \int p(\mathbf{x}, t \mid \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0$ via the continuity equation.

Proof. We follow the standard flow-matching construction [10, 11]: the marginal velocity field is defined by the displayed conditional expectation, and it remains to verify that this definition makes $\mathbf{v}(\mathbf{x}, t)$ generate $p(\mathbf{x}, t)$ via the continuity equation. By the law of total probability and the definition of the conditional velocity field,

$$p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) = \int p(\mathbf{x}, t \mid \mathbf{x}_0) \mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0. \quad (8)$$

Each conditional pair $(\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0), p(\mathbf{x}, t \mid \mathbf{x}_0))$ satisfies the continuity equation $\partial_t p(\mathbf{x}, t \mid \mathbf{x}_0) + \text{div}(p(\mathbf{x}, t \mid \mathbf{x}_0) \mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)) = 0$ by assumption [44]. Integrating against $p(\mathbf{x}_0)$ and exchanging

Algorithm 1 Training the $\beta = 1$ corner instantiation (EMA tangent + FM anchor)

Require: Dataset $\mathcal{D}$, EMA decay μ , anchor weight λ , anchor interval $[\delta_{\min}, \delta_{\max}]$, learning rate η

Initialize $\bar{\theta} \leftarrow \theta$

for $k = 1, 2, \dots$ **do**

Sample $(x_0, x_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $t \sim \mathcal{U}[0, 1]$, $r \sim \mathcal{U}[0, t]$, $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$

$x_t \leftarrow (1-t)x_0 + tx_1$, $v_{\text{cond}} \leftarrow x_1 - x_0$

$v_{\text{tang}} \leftarrow \text{sg}[\mathbf{u}_{\bar{\theta}}(x_t, t, t)]$ $\triangleright$ EMA tangent (one extra forward pass)

$V_{\theta} \leftarrow \mathbf{u}_{\theta}(x_t, r, t) + (t-r) \text{sg}[\text{JVP}(\mathbf{u}_{\theta}, (x_t, r, t), (v_{\text{tang}}, 0, 1))]$

$\ell_{\text{MF}} \leftarrow \|V_{\theta} - v_{\text{cond}}\|_2^2$

$\ell_{\text{FM}} \leftarrow \|\mathbf{u}_{\theta}(x_t, t-\delta, t) - v_{\text{cond}}\|_2^2$ $\triangleright$ FM anchor (one extra forward pass)

$\theta \leftarrow \theta - \eta \nabla_{\theta}(\ell_{\text{MF}} + \lambda \ell_{\text{FM}})$

$\bar{\theta} \leftarrow \mu \bar{\theta} + (1-\mu)\theta$ $\triangleright$ EMA update

end for

return θ $\triangleright$ Generate: $\hat{x}_0 = x_1 - \mathbf{u}_{\theta}(x_1, 0, 1)$

integration with ∂_t and div yields $\partial_t p(x, t) + \text{div}(p(x, t)v(x, t)) = 0$, so the constructed $v(x, t)$ generates $p(x, t)$. Dividing the displayed equation by $p(x, t)$ and applying Bayes' rule $p(x_0 | x_t = x) = p(x, t | x_0)p(x_0)/p(x, t)$ recovers the conditional-expectation form. We treat this as the operating definition of the marginal velocity used throughout; uniqueness up to divergence-free vector fields, which we do not require, is a separate question we do not address. $\square$

D Implementation Details

To validate the framework empirically (Section 5), we instantiate the $\beta = 1$ corner of Theorem 4 as a backbone-preserving training recipe. Two design choices realize the corner at near-zero compute overhead.

EMA proxy. We take $\hat{v} = \mathbf{u}_{\bar{\theta}}(x_t, t, t)$ where $\bar{\theta}$ is a Polyak-averaged copy of θ [45, 46], $\bar{\theta} \leftarrow \mu \bar{\theta} + (1-\mu)\theta$ with $\mu \in [0, 1]$. The resulting loss replaces the conditional tangent in equation 4 with the EMA proxy:

$$\mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) = \mathbb{E}_{r, t, x_0, x_1} \left[\left\| \mathbf{u}_{\theta} + (t-r) \text{sg}[\text{JVP}(\mathbf{u}_{\theta}, (z, r, t), (\mathbf{u}_{\bar{\theta}, t}, 0, 1))] - v_{\text{cond}} \right\|_2^2 \right], \quad (9)$$

with $\mathbf{u}_{\bar{\theta}, t} \triangleq \mathbf{u}(x_t, t, t; \bar{\theta})$. Crucially, the regression target remains v_{cond} : the tangent and target play distinct statistical roles per Section 3.1, and only the tangent needs to be replaced.

Flow-matching anchor. The proxy bias $\mathbf{b} = \mathbf{u}_{\bar{\theta}, t} - v$ must be controlled or it propagates through the $\beta(\mathbf{J} + \mathbf{I})\mathbf{b}$ bias term. We supervise the boundary condition $\mathbf{u}(x_t, t, t) = v(x_t, t)$ directly with a small flow-matching loss

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, x_0, x_1} \left[\left\| \mathbf{u}_{\theta}(x_t, t-\delta, t) - v_{\text{cond}} \right\|_2^2 \right], \quad (10)$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. We chose to evaluate the anchor at a small *interior* time offset rather than exactly at the boundary $\delta = 0$ for two reasons: (i) the same $\mathbf{u}(x_t, t-\delta, t)$ is what the EMA tangent eventually converges to, so the anchor and the JVP target share the same network-evaluation pattern, and (ii) we found training to be slightly more stable when the anchor probes a thin neighborhood of the boundary rather than a single $r = t$ slice. For small δ , $\mathbf{u}(x_t, t-\delta, t)$ approximates the instantaneous velocity, and equation 1 guarantees $\mathbb{E}[v_{\text{cond}} | x_t] = v(x_t, t)$, so $\mathcal{L}_{\text{FM}}$ drives $\mathbf{u}_{\theta}(x_t, t, t) \rightarrow v(x_t, t)$, and via EMA, $\mathbf{u}_{\bar{\theta}}(x_t, t, t) \rightarrow v$ and hence $\mathbf{b} \rightarrow \mathbf{0}$. The full $\beta = 1$ corner loss is

$$\mathcal{L}_{\beta=1}(\theta) = \mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) + \lambda \mathcal{L}_{\text{FM}}(\theta), \quad (11)$$

with $\lambda > 0$. The training procedure is Algorithm 1. Anchor hyperparameters used in our experiments: for 2-D toy and DGMM runs, $(\delta_{\min}, \delta_{\max}, \lambda) = (10^{-4}, 10^{-2}, 0.5)$; for the DiT-B/4 / ImageNet-256 $\beta = 1$ run, $(\delta_{\min}, \delta_{\max}, \lambda) = (0, 10^{-3}, 0.1)$.

917 **Optimality.** The combination of EMA proxy plus FM anchor approximates the $\beta \rightarrow 1$ limit of The-
 918 orem 4 at the cost of one additional JVP per step. Propositions 2 and 3 (proofs in Appendix E)
 919 characterize the resulting gradient and bias structure:

920 **Proposition 2** (Gradient under the $\beta = 1$ corner loss). *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form*

$$\nabla_{\theta} \mathcal{L}_{MF}^{EMA} = 2 \mathbb{E}[g \tilde{r}^{EMA}],$$

921 where $(\nabla_{\theta} \mathbf{u}_{\theta}) \in \mathbb{R}^{d \times p}$ is the parameter Jacobian, $\tilde{r}^{EMA} \triangleq V_{\theta}^{EMA} - \mathbf{v}(\mathbf{x}_t, t) \in \mathbb{R}^d$, and no $\mathbf{J} \mathbf{v}'$ noise
 922 term appears.

923 **Proposition 3** (Bias-Variance Tradeoff). *The EMA mean-field residual decomposes as*

$$\tilde{r}^{EMA} = \mathbf{r}_{\theta} + (t-r) \partial_z \mathbf{u}_{\theta}(\mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)),$$

924 where $\mathbf{r}_{\theta}$ is the true MeanFlow residual. The FM anchor drives $\mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$, giving
 925 $\tilde{r}^{EMA} \rightarrow \mathbf{0}$.

926 **Why the target stays unbiased.** Propositions 2 and 3 treat the proxy $\hat{\mathbf{v}}$ as a tangent-only re-
 927 placement. A natural alternative is to substitute $\hat{\mathbf{v}}$ for the regression target $\mathbf{v}_{\text{cond}}$ as well, yielding a
 928 “fully self-consistent” loss in the spirit of self-distillation. The next proposition shows that the two
 929 replacements are not interchangeable: only the tangent admits a gap-attenuated bias.

930 **Proposition 4** (Target/Tangent Bias Asymmetry). *Let $\hat{\mathbf{v}} = \mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t)$ be a deterministic proxy
 931 with bias $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}(\mathbf{x}_t, t)$. Consider two ways of substituting $\hat{\mathbf{v}}$ into equation 4:*

932 (i) Tangent replacement (the $\beta = 1$ corner): *keep the target $\mathbf{v}_{\text{cond}}$ but replace the JVP tangent.*
 933 *At the loss minimum,*

$$\mathbf{u}^*(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t) = -\frac{1}{2}(t-r) \partial_z \mathbf{u}^* \mathbf{b} + \mathcal{O}((t-r) \|\mathbf{b}\|^2),$$

934 *so the stationary error is $\mathcal{O}((t-r) \|\partial_z \mathbf{u}\| \|\mathbf{b}\|)$ and vanishes at $r \rightarrow t$.*

935 (ii) Target replacement: *substitute $\hat{\mathbf{v}}$ for $\mathbf{v}_{\text{cond}}$ in the regression target. At $r = t$ the loss minimum*
 936 *satisfies*

$$\mathbf{u}^*(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t) = \mathbf{b},$$

937 *i.e., the stationary error equals the proxy bias exactly, uniformly in t and independent of*
 938 *the gap $(t-r)$.*

939 The proof is given in Appendix E. The asymmetry has three immediate consequences. *First*, the
 940 $\beta = 1$ corner’s design choice (replace only the tangent, keep $\mathbf{v}_{\text{cond}}$ as target) is not arbitrary: tangent-
 941 replacement bias is multiplicatively damped by the gap $(t-r)$, while target-replacement bias is not.
 942 *Second*, the FM anchor in equation 10 suffices to control the residual tangent-replacement bias
 943 because it pins $\mathbf{u}_{\theta}(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}$ at the boundary $r = t$, where any target-replacement scheme would
 944 require an analogous anchor at *every* (r, t) . *Third*, the proposition locates a failure mode in self-
 945 distillation-style MeanFlow variants that use a deterministic proxy for both roles: the loss admits
 946 the trivial fixed point $\mathbf{u}^* \equiv \hat{\mathbf{v}}$, which is preserved under proxy updates and only escaped by external
 947 supervision pulling $\hat{\mathbf{v}}$ back toward $\mathbf{v}$ at every t . Concurrent methods that adopt this pattern indeed
 948 report sensitivity to the proxy-update schedule and the boundary regularizer [23, 24].

949 Proposition 2 confirms that the $\mathbf{J} \mathbf{v}'$ amplification of Theorem 2 is eliminated; Proposition 3 confirms
 950 that the residual EMA bias is controlled by the FM anchor, recovering the asymptotic optimum
 951 of Theorem 4; Proposition 4 explains why the FM anchor at the boundary $r = t$ alone is sufficient.

952 E Proofs of Theorems and Propositions

953 E.1 Proof of Theorem 2 (Jacobian Variance Amplification)

954 **Theorem 5** (Jacobian Variance Amplification). *The Jacobi factor amplifies the gradient variance*
 955 *of the original MeanFlow loss:*

$$\text{Var}[\nabla_{\theta} \ell_{MF}(\theta) \mid \mathbf{x}_t] \propto \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}).$$

956 *Proof.* Under the stop-gradient operator, the compound prediction V_θ^{sg} depends on θ only through
 957 $\mathbf{u}_\theta(\mathbf{x}_t, r, t)$. Let $\nabla_\theta \mathbf{u}_\theta \in \mathbb{R}^{d \times p}$ denote the parameter Jacobian of the network output. The per-step
 958 gradient is

$$\nabla_\theta \ell_{\text{MF}} = 2(\nabla_\theta \mathbf{u}_\theta)^\top (\mathbf{r}_\theta^{\text{sg}} + \mathbf{J} \mathbf{v}'), \quad (12)$$

959 where $\mathbf{r}_\theta^{\text{sg}}$ is the mean-field residual and $\mathbf{J} = (t - r) \partial_z \mathbf{u}_\theta - \mathbf{I}_d$ is the Jacobi factor. All three
 960 quantities ($\nabla_\theta \mathbf{u}_\theta$, $\mathbf{r}_\theta^{\text{sg}}$, and $\mathbf{J}$) are deterministic conditioned on $\mathbf{x}_t$.

961 Since $\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$ by equation 1, the conditional mean gradient is:

$$\mathbb{E}[\nabla_\theta \ell | \mathbf{x}_t] = 2(\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{r}_\theta^{\text{sg}}. \quad (13)$$

962 The centered gradient is $\nabla_\theta \ell - \mathbb{E}[\nabla_\theta \ell | \mathbf{x}_t] = 2(\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{J} \mathbf{v}'$, so the conditional variance is:

$$\begin{aligned} \text{Var}[\nabla_\theta \ell | \mathbf{x}_t] &= \mathbb{E}[\|2(\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{J} \mathbf{v}'\|^2 | \mathbf{x}_t] \\ &= 4 \mathbb{E} \left[(\mathbf{v}')^\top \mathbf{J}^\top \underbrace{(\nabla_\theta \mathbf{u}_\theta)(\nabla_\theta \mathbf{u}_\theta)^\top}_{\mathbf{G}_\theta \succeq \mathbf{0}} \mathbf{J} \mathbf{v}' \mid \mathbf{x}_t \right] \\ &= 4 \text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top) \propto \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top), \end{aligned} \quad (14)$$

963 where $\mathbf{G}_\theta = (\nabla_\theta \mathbf{u}_\theta)(\nabla_\theta \mathbf{u}_\theta)^\top \in \mathbb{R}^{d \times d}$ is the parameter-space Gram matrix and the last step uses
 964 $\mathbf{G}_\theta \succeq \mathbf{0}$ to establish proportionality via the cyclic property of the trace. $\square$

965 E.2 Proof of Theorem 3 (Semi-Gradient Gap)

966 **Theorem 6** (Semi-Gradient Gap). *The gradients of the MeanFlow loss with and without the stop-*
 967 *gradient operator differ by*

$$\underbrace{2(t-r) \mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} [(\nabla_\theta(\partial_z \mathbf{u}_\theta \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_\theta))^\top \mathbf{r}_\theta]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} [\nabla_\theta \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)]}_{\text{variance-driven gradient difference}}.$$

968 *Proof.* The MeanFlow loss conditioned on $\mathbf{x}_t$ decomposes as (cf. equation 5):

$$\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\ell_{\text{MF}}] = \|\mathbf{r}_\theta^{\text{sg}}\|^2 + \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top). \quad (15)$$

969 **Without stop-gradient.** Both terms depend on θ (through $\mathbf{u}_\theta$ and $\partial_z \mathbf{u}_\theta$), so the full gradient is:

$$\nabla_\theta \mathbb{E}[\ell] = \nabla_\theta \|\mathbf{r}_\theta\|^2 + \nabla_\theta \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top). \quad (16)$$

970 **With stop-gradient.** The JVP terms in $\mathbf{r}_\theta^{\text{sg}}$ and $\mathbf{J}$ are treated as constants, so:

$$\nabla_\theta^{\text{sg}} \|\mathbf{r}_\theta^{\text{sg}}\|^2 = 2(\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{r}_\theta^{\text{sg}}, \quad \nabla_\theta^{\text{sg}} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top) = \mathbf{0}. \quad (17)$$

971 The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned} \nabla_\theta \mathbb{E}[\ell] - \nabla_\theta^{\text{sg}} \mathbb{E}[\ell] &= \underbrace{\nabla_\theta \|\mathbf{r}_\theta\|^2 - 2(\nabla_\theta \mathbf{u}_\theta)^\top \mathbf{r}_\theta^{\text{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_\theta \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)}_{\text{variance-driven difference}}. \end{aligned} \quad (18)$$

972 Expanding the first term: $\nabla_\theta \|\mathbf{r}_\theta\|^2 = 2(\nabla_\theta \mathbf{r}_\theta)^\top \mathbf{r}_\theta$, and $\nabla_\theta \mathbf{r}_\theta$ includes the derivative of the JVP
 973 terms $(t - r)(\partial_z \mathbf{u}_\theta \cdot \mathbf{v} + \partial_t \mathbf{u}_\theta)$ w.r.t. θ , yielding the second-order mean-field difference $2(t -$
 974 $r) \mathbb{E}[(\nabla_\theta(\partial_z \mathbf{u}_\theta \cdot \mathbf{v} + \partial_t \mathbf{u}_\theta))^\top \mathbf{r}_\theta]$. At convergence $\mathbf{r}_\theta \rightarrow \mathbf{0}$, this term vanishes, and the gap is
 975 dominated by the variance-driven term $\nabla_\theta \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)$. $\square$

976 E.3 Proof of Proposition 2 (Gradient under the $\beta = 1$ corner loss)

977 **Proposition 7** (Gradient under the $\beta = 1$ corner loss). *The gradient of $\mathcal{L}_{\text{MF}}^{\text{EMA}}$ takes the form*

$$\nabla_\theta \mathcal{L}_{\text{MF}}^{\text{EMA}} = 2 \mathbb{E}[(\nabla_\theta \mathbf{u}_\theta(\mathbf{x}_t, r, t))^\top \tilde{\mathbf{r}}^{\text{EMA}}],$$

978 where $\tilde{\mathbf{r}}^{\text{EMA}} := V_\theta^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t)$ is the mean-field residual under the EMA tangent, and no $\mathbf{J} \mathbf{v}'$ noise
 979 term appears.

980 *Proof.* The EMA velocity anchor $\mathbf{u}_\theta(\mathbf{x}_t, t, t)$ is deterministic given $\mathbf{x}_t$, since the EMA parameters θ are fixed during the gradient step. Under stop-gradient, V_θ^{EMA} depends on θ only through $\mathbf{u}_\theta(\mathbf{x}_t, r, t)$. By the same argument as equation 12, the per-step gradient is:

$$\nabla_\theta \ell = 2 (\nabla_\theta \mathbf{u}_\theta)^\top (V_\theta^{\text{EMA}} - \mathbf{v}_{\text{cond}}). \quad (19)$$

983 Writing $\mathbf{v}_{\text{cond}} = \mathbf{v}(\mathbf{x}_t, t) + \mathbf{v}'$ and noting that $\tilde{\mathbf{r}}^{\text{EMA}} := V_\theta^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t)$ is deterministic given $\mathbf{x}_t$:

$$\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\nabla_\theta \ell] = 2 (\nabla_\theta \mathbf{u}_\theta)^\top \tilde{\mathbf{r}}^{\text{EMA}}, \quad (20)$$

984 since $\mathbb{E}[\mathbf{v}' | \mathbf{x}_t] = \mathbf{0}$. The tower property $\nabla_\theta \mathcal{L}_{\text{MF}}^{\text{EMA}} = \mathbb{E}_{r,t,\mathbf{x}_t}[\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\nabla_\theta \ell]]$ gives the stated result. Crucially, no $\mathbf{J}\mathbf{v}'$ term appears because the EMA tangent is deterministic; compare with the original MeanFlow gradient equation 12 where the stochastic tangent produces the $\mathbf{J}\mathbf{v}'$ noise. $\square$

987 E.4 Proof of Proposition 3 (Bias-Variance Tradeoff)

988 **Proposition 8** (Bias-Variance Tradeoff). *The EMA mean-field residual decomposes as*

$$\tilde{\mathbf{r}}^{\text{EMA}} = \mathbf{r}_\theta + (t-r) \partial_z \mathbf{u}_\theta(\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)),$$

989 where $\mathbf{r}_\theta$ is the true MeanFlow residual from equation 5. At convergence $\mathbf{r}_\theta \rightarrow \mathbf{0}$ and the FM anchor drives $\mathbf{u}_\theta(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$, giving $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$.

991 *Proof.* Expanding V_θ^{EMA} from equation 9, the value of the compound prediction (ignoring the stop-gradient, which does not affect values) is:

$$V_\theta^{\text{EMA}} = \mathbf{u}_\theta + (t-r)(\partial_z \mathbf{u}_\theta \cdot \mathbf{u}_\theta(\mathbf{x}_t, t, t) + \partial_t \mathbf{u}_\theta). \quad (21)$$

993 The true MeanFlow residual (with the marginal velocity as tangent) is:

$$\mathbf{r}_\theta = \mathbf{u}_\theta + (t-r)(\partial_z \mathbf{u}_\theta \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_\theta) - \mathbf{v}(\mathbf{x}_t, t). \quad (22)$$

994 Subtracting:

$$\begin{aligned} \tilde{\mathbf{r}}^{\text{EMA}} &= V_\theta^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t) \\ &= \mathbf{r}_\theta + (t-r) \partial_z \mathbf{u}_\theta(\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)). \end{aligned} \quad (23)$$

995 At convergence, the true residual $\mathbf{r}_\theta \rightarrow \mathbf{0}$ (the MeanFlow identity is satisfied) and the FM anchor loss equation 10 supervises the boundary condition, driving $\mathbf{u}_\theta(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$ and hence $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$. $\square$

998 E.5 Proof of Proposition 4 (Target/Tangent Bias Asymmetry)

999 **Proposition 9** (Target/Tangent Bias Asymmetry). *Let $\hat{\mathbf{v}} = \mathbf{u}_\theta(\mathbf{x}_t, t, t)$ be a deterministic proxy with bias $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}(\mathbf{x}_t, t)$. For the tangent-replacement loss (the $\beta = 1$ corner), the loss minimum $\mathbf{u}^*$ satisfies $\mathbf{u}^*(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t) = -\frac{1}{2}(t-r) \partial_z \mathbf{u}^* \mathbf{b} + \mathcal{O}((t-r) \|\mathbf{b}\|^2)$. For the target-replacement loss, the loss minimum at $r=t$ satisfies $\mathbf{u}^*(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t) = \mathbf{b}$.*

1003 *Proof. Tangent replacement.* Substituting $\hat{\mathbf{v}}$ for $\mathbf{v}_{\text{cond}}$ only in the JVP tangent (target unchanged) gives the per-sample compound prediction

$$V_\theta^{\text{EMA}} = \mathbf{u}_\theta + (t-r)(\partial_z \mathbf{u}_\theta \cdot \hat{\mathbf{v}} + \partial_t \mathbf{u}_\theta),$$

1005 and the loss is $\mathbb{E}[\|V_\theta^{\text{EMA}} - \mathbf{v}_{\text{cond}}\|^2]$. By stationarity, $\mathbb{E}[V_\theta^{\text{EMA}} - \mathbf{v}_{\text{cond}} | \mathbf{x}_t] = \mathbf{0}$ at the minimum. Taking conditional expectation and using $\mathbb{E}[\mathbf{v}_{\text{cond}} | \mathbf{x}_t] = \mathbf{v}(\mathbf{x}_t, t)$,

$$\mathbf{u}^* + (t-r)(\partial_z \mathbf{u}^* \cdot \hat{\mathbf{v}} + \partial_t \mathbf{u}^*) = \mathbf{v}(\mathbf{x}_t, t).$$

1007 At the same point, the *true* MeanFlow identity (with the marginal velocity as tangent) reads

$$\mathbf{u}^\diamond + (t-r)(\partial_z \mathbf{u}^\diamond \cdot \mathbf{v} + \partial_t \mathbf{u}^\diamond) = \mathbf{v}(\mathbf{x}_t, t),$$

1008 with $\mathbf{u}^\diamond(\mathbf{x}_t, t, t) = \mathbf{v}(\mathbf{x}_t, t)$. Subtracting and writing $\mathbf{u}^* = \mathbf{u}^\diamond + \Delta$,

$$\Delta + (t-r) \partial_z \mathbf{u}^* \mathbf{b} + (t-r) \text{D}_t \Delta = \mathbf{0}, \quad \text{D}_t \Delta \triangleq \partial_t \Delta + \mathbf{v} \cdot \partial_z \Delta,$$

with the boundary condition $\Delta(\mathbf{x}_t, t, t) = \mathbf{0}$. Substituting $s = t - r$ at fixed r converts this to a first-order linear ODE in s :

$$s \frac{d\Delta}{ds} + \Delta = -s \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + \mathcal{O}(\|\mathbf{b}\|^2),$$

which has $s\Delta(s) = -\frac{s^2}{2} \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + C$; the boundary condition $\Delta(0) = \mathbf{0}$ forces $C = 0$. Hence

$$\Delta(\mathbf{x}_t, r, t) = -\frac{1}{2}(t - r) \partial_{\mathbf{z}} \mathbf{u}^* \mathbf{b} + \mathcal{O}((t - r) \|\mathbf{b}\|^2),$$

i.e., the stationary error is $\mathcal{O}((t - r) \|\partial_{\mathbf{z}} \mathbf{u}\| \|\mathbf{b}\|)$ and vanishes at the boundary $r \rightarrow t$. The factor of $\frac{1}{2}$ comes from integrating the material-derivative term, which contributes equally to the leading-order coefficient as the bias source term.

Target replacement. Substituting $\hat{\mathbf{v}}$ for $\mathbf{v}_{\text{cond}}$ in the regression target yields the loss $\mathbb{E} \|\mathbf{u}_{\theta} + (t - r) D\mathbf{u}/Dt - \hat{\mathbf{v}}\|^2$. By stationarity, the loss minimum satisfies $\mathbb{E}[\mathbf{u}^* + (t - r) D\mathbf{u}^*/Dt - \hat{\mathbf{v}} \mid \mathbf{x}_t] = \mathbf{0}$. Specializing to $r = t$, the gap term vanishes and the equation collapses to $\mathbf{u}^*(\mathbf{x}_t, t, t) = \hat{\mathbf{v}} = \mathbf{v}(\mathbf{x}_t, t) + \mathbf{b}$, so the stationary error at the boundary is exactly $\mathbf{b}$, independent of (t, r) .

Asymmetry. Tangent replacement carries a multiplicative $(t - r)$ factor on the bias and is therefore pinned to zero at the boundary $r = t$ that the FM anchor enforces; target replacement carries no such factor and the bias persists at every t , requiring a boundary regularizer at *every* (r, t) to remove. $\square$

E.6 Proof of Theorem 4 (Optimal Control-Variate Coefficient)

Theorem 10 (Optimal control-variate coefficient). *Under the scalar-isotropic approximation $\Sigma_{\mathbf{v}'} \approx \sigma^2 \mathbf{I}_d$ and $\mathbf{J} \approx \kappa \mathbf{I}_d$, the conditional MSE of $\mathbf{g}^{(\beta)}$ in equation 7 is*

$$M(\beta) = \beta^2 (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1 - \beta)\kappa - \beta)^2,$$

with unique minimizer $\beta^* = \frac{\kappa}{\kappa + 1} \cdot \frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}$ and optimum value $M(\beta^*) = \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$. For all κ , $\|\mathbf{b}\|^2 > 0$, the optimum strictly dominates both $M(0) = \sigma^2 \kappa^2 d$ and $M(1) = (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d$.

Proof. Step 1: bias-variance decomposition. From equation 7,

$$\mathbf{g}^{(\beta)} = 2\mathbf{g}^\top \left[\bar{\mathbf{r}} + ((1 - \beta)\mathbf{J} - \beta\mathbf{I}) \mathbf{v}' + \beta(\mathbf{J} + \mathbf{I}) \mathbf{b} \right].$$

Using $\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$, the conditional mean is $\mathbb{E}[\mathbf{g}^{(\beta)} \mid \mathbf{x}_t] = 2\mathbf{g}^\top (\bar{\mathbf{r}} + \beta(\mathbf{J} + \mathbf{I}) \mathbf{b})$ and the centered noise is $-2\mathbf{g}^\top ((1 - \beta)\mathbf{J} - \beta\mathbf{I}) \mathbf{v}'$ (relative to the ideal target $2\mathbf{g}^\top \bar{\mathbf{r}}$). The conditional MSE w.r.t. the ideal target therefore decomposes as

$$\text{MSE} = \underbrace{4\beta^2 \|\mathbf{g}^\top (\mathbf{J} + \mathbf{I}) \mathbf{b}\|^2}_{\text{bias}^2} + \underbrace{4 \text{Tr}(\mathbf{g} \mathbf{g}^\top ((1 - \beta)\mathbf{J} - \beta\mathbf{I}) \Sigma_{\mathbf{v}'} ((1 - \beta)\mathbf{J} - \beta\mathbf{I})^\top)}_{\text{variance}}.$$

Step 2: matrix-form expansion and strict convexity. Let $\mathbf{A} \triangleq \mathbf{J} + \mathbf{I}$ and $\mathbf{G}_\theta \triangleq \mathbf{g} \mathbf{g}^\top \succeq \mathbf{0}$. Substituting $((1 - \beta)\mathbf{J} - \beta\mathbf{I}) = \mathbf{J} - \beta\mathbf{A}$ and using $\|\mathbf{g}^\top \mathbf{A} \mathbf{b}\|^2 = \text{Tr}(\mathbf{G}_\theta \mathbf{A} \mathbf{b} \mathbf{b}^\top \mathbf{A}^\top)$, the conditional MSE writes as a quadratic in β :

$$M(\beta) = \alpha_0 - 2\alpha_1 \beta + \alpha_2 \beta^2, \tag{24}$$

with (suppressing the constant prefactor 4)

$$\begin{aligned} \alpha_0 &\triangleq \text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top), \\ \alpha_1 &\triangleq \text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{A}^\top), \\ \alpha_2 &\triangleq \text{Tr}(\mathbf{G}_\theta \mathbf{A} (\Sigma_{\mathbf{v}'} + \mathbf{b} \mathbf{b}^\top) \mathbf{A}^\top). \end{aligned}$$

(We used the symmetry of $\mathbf{G}_\theta$ and $\Sigma_{\mathbf{v}'}$ to consolidate the linear-in- β trace into a single α_1 .) The Hessian is

$$\frac{\partial^2 M}{\partial \beta^2} = 2\alpha_2 = 2 \text{Tr}(\mathbf{G}_\theta \mathbf{A} (\Sigma_{\mathbf{v}'} + \mathbf{b} \mathbf{b}^\top) \mathbf{A}^\top) \geq 0,$$

since each factor inside the trace is positive semi-definite ($\mathbf{G}_\theta \succeq \mathbf{0}$, $\Sigma_{\mathbf{v}'} + \mathbf{b} \mathbf{b}^\top \succeq \mathbf{0}$, and $\mathbf{A}(\Sigma_{\mathbf{v}'} + \mathbf{b} \mathbf{b}^\top) \mathbf{A}^\top = \mathbf{M} \mathbf{M}^\top \succeq \mathbf{0}$ for $\mathbf{M} \triangleq \mathbf{A}(\Sigma_{\mathbf{v}'} + \mathbf{b} \mathbf{b}^\top)^{1/2}$). Hence $M(\beta)$ is convex in β . The Hessian is

strictly positive whenever $\mathbf{G}_\theta^{1/2} \mathbf{M} \neq \mathbf{0}$, which holds outside two pathological regimes: (i) $\mathbf{A} = \mathbf{0}$, equivalently $\mathbf{J} = -\mathbf{I}$ (network with $\partial_z \mathbf{u}_\theta \equiv \mathbf{0}$, e.g., at saturated initialization), in which case the linear coefficient α_1 also vanishes and $M(\beta)$ is constant; (ii) $\mathbf{g} = \mathbf{0}$ (frozen parameters). Outside these, $M(\beta)$ is *strictly* convex with unique unconstrained minimizer

$$\beta_{\text{matrix}}^* = \frac{\alpha_1}{\alpha_2} = \frac{\text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} (\mathbf{J} + \mathbf{I})^\top)}{\text{Tr}(\mathbf{G}_\theta (\mathbf{J} + \mathbf{I}) (\Sigma_{\mathbf{v}'} + \mathbf{b} \mathbf{b}^\top) (\mathbf{J} + \mathbf{I})^\top)}. \quad (25)$$

Note that β_{matrix}^* may fall outside $[0, 1]$; the box-constrained optimum on the $\beta \in [0, 1]$ interval (cf. equation 6) is then $\text{clip}(\beta_{\text{matrix}}^*, 0, 1)$, attained at the nearer endpoint by convexity.

Step 3: scalar approximation. Substituting $\mathbf{J} = \kappa \mathbf{I}_d$, $\Sigma_{\mathbf{v}'} = \sigma^2 \mathbf{I}_d$, $\mathbf{A} = (\kappa + 1) \mathbf{I}_d$, and the parameter-isotropy approximation $\mathbf{G}_\theta \propto \mathbf{I}_d$ (which absorbs the parameter-direction normalization into a per-component MSE),

$$\alpha_0 \propto \sigma^2 \kappa^2 d, \quad \alpha_1 \propto \sigma^2 \kappa (\kappa + 1) d, \quad \alpha_2 \propto (\kappa + 1)^2 (\sigma^2 d + \|\mathbf{b}\|^2),$$

which equivalently gives

$$M(\beta) \propto \beta^2 (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1 - \beta) \kappa - \beta)^2.$$

Step 4: minimization. Differentiating and setting to zero (or directly evaluating α_1/α_2),

$$\beta^* = \frac{\alpha_1}{\alpha_2} = \frac{\sigma^2 \kappa (\kappa + 1) d}{(\kappa + 1)^2 (\sigma^2 d + \|\mathbf{b}\|^2)} = \frac{\kappa}{\kappa + 1} \cdot \frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}.$$

Step 5: optimum value and corner dominance. At β^* , $(1 - \beta^*) \kappa - \beta^* = \kappa \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$. Substituting back,

$$M(\beta^*) \propto \frac{\sigma^2 d \kappa^2 \|\mathbf{b}\|^2}{\sigma^2 d + \|\mathbf{b}\|^2}.$$

For $\kappa > 0$ and $\|\mathbf{b}\|^2 > 0$: $M(\beta^*)/M(0) = \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2) < 1$, so $M(\beta^*) < M(0)$. For the $M(1)$ comparison, $M(1) - M(\beta^*) \propto (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d - \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$, and the first two terms exceed $\sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$ since $(\kappa + 1)^2 > \kappa^2$ for $\kappa > 0$. Strict dominance is therefore over the unconstrained interior optimum; if $\beta_{\text{matrix}}^* \notin [0, 1]$ the box-constrained optimum is at the nearer corner and the dominance is non-strict. $\square$

F DiT Latent-Space Training on ImageNet

We test whether the variance-reduction mechanism of Theorems 2 and 4 scales beyond toys by training DiT [42] variants on ImageNet in the latent space of a frozen Stable Diffusion VAE, following the recipe of [21]. The four β configurations share identical hyperparameters and code paths; only the loss differs (vanilla MeanFlow MSE for the interior β runs versus the EMA-tangent loss in equation 11 for the $\beta = 1$ corner).

F.1 DiT-B/4, ImageNet-256 $\times$ 256

Setup: logit-normal (r, t) sampling with mean -0.4 and stddev 1, overlap rate 0.75, AdamW with constant learning rate 10^{-4} , EMA decay 0.9999, batch size 256 across 32 TPU v4 chips. All four configurations ($\beta \in \{0, 0.25, 0.5, 1\}$) share the same DiT-B/4 backbone, data pipeline, optimizer, EMA schedule, and (r, t) sampling distribution. The interior β runs ($\beta \in \{0, 0.25, 0.5\}$) reuse the baseline configuration verbatim and add only the tangent-mixing rule of equation 6; the $\beta = 1$ corner additionally disables the Karras-style adaptive loss weighting and adds the FM anchor of equation 10 ($\delta \sim \mathcal{U}[0, 10^{-3}]$, $\lambda = 0.1$). The extra EMA forward pass and FM-anchor evaluation per step add up to a $\sim 22\%$ wall-clock overhead on TPU v4 (2.20 steps/s for $\beta = 1$ versus 2.65 steps/s for the baseline); the $\beta \in \{0.25, 0.5\}$ interior runs incur only the EMA forward overhead ($\sim 10\%$).

Four-point matched-step ordering. Table 5 reports FID_{50k} at representative 5k-step checkpoints for all four runs. After the early-training transient ($\leq 25k$ steps), the empirical four-point ordering $\text{FID}(\beta = 0) < \text{FID}(\beta = 0.25) < \text{FID}(\beta = 0.5) < \text{FID}(\beta = 1)$ holds at *every* 5k-aligned matched-step checkpoint we logged (54 consecutive datapoints from step 30k to step 295k). The $\beta = 0.25$

vs baseline gap stays positive throughout, ranging from +9.5 early in training (step 30k) to a tight +0.39 at step 295k. The $\beta = 0.5$ vs baseline gap stabilizes around +1.0 $\sim$ +1.5 in the converged regime (steps 200k to 295k), and the $\beta = 1$ vs baseline gap remains +12 $\sim$ +14 across the converged regime.

Converged FID floors and four-point ordering. All four configurations completed their scheduled 300k-step training and have a matched-step eval at step 295k (the baseline crashed at step 239,950 and was resumed from its 240k checkpoint to reach the same horizon as the other three). The four FID floors at step 295k recover the bias-variance ordering of Theorem 4:

$$\text{FID}^*(\beta=0) = 11.37 < \text{FID}^*(\beta=0.25) = 11.76 < \text{FID}^*(\beta=0.5) = 12.51 < \text{FID}^*(\beta=1) = 23.36.$$

Quantitatively, taking offsets from the baseline floor 11.37, $\Delta\text{FID}(\beta=0.5) \approx 1.14$ and $\Delta\text{FID}(\beta=1) \approx 11.99$. Anchoring the gradient-MSE prediction $\beta^2 \cdot A$ at the $\beta=0.5$ point gives $A \approx 4.58$, so the predicted offset at $\beta=1$ is 4.58 — the empirical 11.99 exceeds this by $\sim 2.6\times$. The FID landscape is therefore *super-linear* in the MSE-axis bias: it penalizes large bias more aggressively than gradient MSE does, which pulls the FID-optimal β past the gradient-MSE interior minimizer to the unbiased corner $\beta=0$. The four-point ordering is the strongest empirical confirmation of the bias-variance decomposition we obtain at DiT scale.

Table 5: DiT-B/4 / ImageNet-256 FID_{50k} (lower is better) at representative 5k-step checkpoints across the four-point β -sweep. All four runs completed their scheduled 300k-step training and have a matched-step eval at step 295k (the baseline crashed at step 239,950 and was resumed from its 240k checkpoint to reach 295k). Bottom rows give the matched-step deltas. The four-point ordering $\text{FID}(\beta=0) < \text{FID}(\beta=0.25) < \text{FID}(\beta=0.5) < \text{FID}(\beta=1)$ holds at every 5k-aligned matched-step checkpoint after the early-training transient (54 consecutive datapoints from step 30k to step 295k).

step (k)	30	50	100	150	200	250	275	295
baseline ($\beta=0$)	128.9	54.5	22.0	15.5	13.0	11.9	11.6	11.37
$\beta=0.25$	138.5	61.6	23.3	16.2	13.3	12.1	11.7	11.76
$\beta=0.5$	145.2	66.9	25.6	17.6	14.4	13.2	12.7	12.51
$\beta=1$	148.8	79.9	41.7	32.1	27.6	25.1	24.1	23.36
$\Delta(\beta=0.25 - \text{baseline})$	+9.5	+7.1	+1.3	+0.7	+0.4	+0.2	+0.1	+0.39
$\Delta(\beta=0.5 - \text{baseline})$	+16.2	+12.4	+3.6	+2.1	+1.4	+1.3	+1.0	+1.14
$\Delta(\beta=1 - \text{baseline})$	+19.9	+25.4	+19.7	+16.6	+14.6	+13.2	+12.5	+11.99

Bias and shrinkage trajectory. To localize the predicted optimum we measure two diagnostics on the DiT checkpoints: an EMA-tracking proxy $\|\widehat{\mathbf{b}}\|^2 \triangleq \mathbb{E}_{\mathbf{x}_t} [\|\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{u}_{\widehat{\theta}}(\mathbf{x}_t, t, t)\|^2]$ (the boundary gap between current parameters and the EMA copy), and the irreducible noise floor $\sigma^2 d \approx 8.2 \times 10^3$ from $\text{Tr}(\Sigma_{v'})$ on the same batch. Table 6 reports the trajectory: the EMA-tracking proxy *decays* through training, and the shrinkage factor $\sigma^2 d / (\sigma^2 d + \|\widehat{\mathbf{b}}\|^2)$ stays near 1 throughout converged training. We caveat that $\|\widehat{\mathbf{b}}\|^2$ is the EMA-tracking gap, not the true model bias $\|\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)\|^2$; cleanly estimating the latter requires access to the conditional distribution $p(\mathbf{x}_0 | \mathbf{x}_t)$, which is intractable on ImageNet. Combined with $\kappa \approx 1$ from the Frobenius-norm Hutchinson trace, the scalar-isotropic substitution gives $\beta^* \in [0.46, 0.50]$ across the full trajectory; this is a coarse first-cut prediction that the direct matrix-form measurement (Section 5.4) refines to $\beta_{\text{no bias}}^* \approx 0.94$.

Table 6: Direct DiT-checkpoint measurement of the bias-variance decomposition ingredients of Theorem 4 at $t = 0.5$. The EMA bias $\|\mathbf{b}\|^2$ averages over 128 synthetic-Gaussian $\mathbf{x}_t$ samples and is reported for both methods (which track within 20% of each other). The shrinkage factor uses $\sigma^2 d \approx 8.2 \times 10^3$. The predicted β^* uses $\kappa = 1$ from the Hutchinson trace.

step (k)	$\ \mathbf{b}\ ^2$ (avg)	$\sigma^2 d / (\sigma^2 d + \ \mathbf{b}\ ^2)$	β^*
20	~ 700	0.92	0.46
40	~ 150	0.98	0.49
80	~ 50	0.99	0.50